MATH 551: NUMERICAL ORDINARY DIFFERENTIAL EQUATIONS

RUIZHE QIAN

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1. August 24, 2026

1.1. **Ordinary differential equations.** An ODE has the form

$$x'(t) = f(t, x(t)).$$

Given an initial condition $x(t_0) = x_0$, the corresponding *initial-value problem* (IVP) is to find $x(t)$ for $t \geq t_0$ such that

$$x'(t) = f(t, x(t)), \qquad x(t_0) = x_0. \tag{1}$$

Topics arising in the numerical solution of differential equations include

- non-stiff problems and stiff solvers;
- geometric solvers;
- parameter estimation and Bayesian approaches, where the vector field may depend on a model parameter θ, as in $f(t, x, \theta)$;
- boundary-value problems; and
- stochastic differential equations (SDEs).

We generally consider vector-valued solutions and vector fields,

$$x \colon \mathbb{R} \longrightarrow \mathbb{R}^n, \qquad f \colon \mathbb{R} \times \mathbb{R}^n \longrightarrow \mathbb{R}^n,$$

which can be written componentwise as

$$x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \qquad f(t, x) = \begin{bmatrix} f_1(t, x_1, \ldots, x_n) \\ \vdots \\ f_n(t, x_1, \ldots, x_n) \end{bmatrix}.$$

1.2. **Examples.**

Example 1. *The scalar equation*

$$x' = x$$

is an ODE with $n = 1$.

Example 2 (Damped oscillator)**.** *Consider the second-order equation*

$$x'' = -\omega^2 x - x'.$$

Setting $x_1 = x$ *and* $x_2 = x'$ *converts it into the first-order system*

$$\begin{cases} x_1' = x_2, \\ x_2' = -\omega^2 x_1 - x_2. \end{cases}$$

Thus, with $\mathbf{x} = (x_1, x_2)^{\mathsf{T}}$*, we have* $\mathbf{x}' = \mathbf{f}(x_1, x_2)$.

Example 3 (Chemical kinetics)**.** *A three-species kinetic model is*

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \qquad \begin{cases} x_1' = -k_1 x_1 + k_2 x_2 x_3, \\ x_2' = k_1 x_1 - k_2 x_2 x_3 - k_3 x_2^2, \\ x_3' = k_3 x_2^2, \end{cases}$$

where $k_1, k_2, k_3 > 0$ are reaction rates. The reaction rates may occur on very different scales, which can lead to stiffness.

Example 4 (Molecular model)**.** *Let q_i be the position of particle i, let m_i be its mass, and let $V(q_1, \ldots, q_N)$ be the potential energy. Newton's equation is*

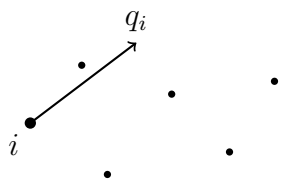
$$m_i q_i'' = -\frac{\partial V(q_1, \ldots, q_N)}{\partial q_i}.$$

Introducing the momentum $p_i = m_i q_i'$ gives

$$\begin{cases} q_i' = \dfrac{p_i}{m_i}, \\ p_i' = -\dfrac{\partial V}{\partial q_i}. \end{cases}$$

The positions and momenta can therefore be collected into the state vector

$$\mathbf{x} = \begin{bmatrix} q_1 \\ p_1 \\ q_2 \\ p_2 \\ \vdots \\ q_N \\ p_N \end{bmatrix}.$$



1.3. **Solutions and the Lipschitz condition.** Consider the IVP (1) on an interval containing t_0.

Definition 1. *A function $x(t)$ is a* solution *of the IVP if it satisfies the integral equation*

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds.$$

Definition 2 (Lipschitz condition)**.** *The vector field f is* Lipschitz continuous in x *on a domain D if there is a constant $L \geq 0$ such that*

$$\|f(t, x) - f(t, y)\| \leq L\|x - y\|$$

for all $x, y \in D$ and all relevant t.

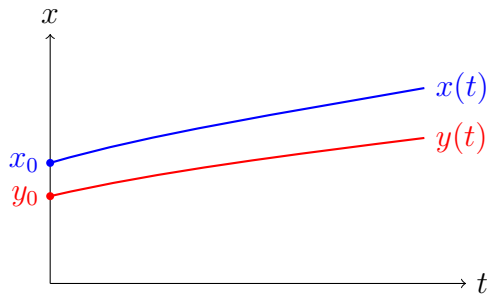
Under the usual continuity assumption in t, the Lipschitz condition in x implies that the solution of the IVP exists and is unique on an appropriate interval.

The same condition also controls the sensitivity to the initial data. Let $x(t)$ and $y(t)$ be solutions with initial values x_0 and y_0, respectively. Their integral equations give

$$\begin{aligned}\|x(t) - y(t)\| &\leq \|x_0 - y_0\| + \int_{t_0}^{t} \|f(s, x(s)) - f(s, y(s))\|\, ds \\ &\leq \|x_0 - y_0\| + L \int_{t_0}^{t} \|x(s) - y(s)\|\, ds.\end{aligned}$$

By Grönwall's inequality,

$$\|x(t) - y(t)\| \leq \|x_0 - y_0\| e^{L(t-t_0)}. \tag{2}$$



Thus nearby initial values produce solutions whose separation is bounded by (2).

2. August 26, 2026

The main topics are properties of solutions, time discretization, and consistency error. We continue to consider the IVP

$$x'(t) = f(t, x(t)), \qquad x(t_0) = x_0, \tag{3}$$

where $x\colon \mathbb{R} \to \mathbb{R}^n$.

2.1. **Stability of solutions.**

The integral formulation of (3) is

$$x(t) = x_0 + \int_{t_0}^{t} f(s, x(s))\, ds.$$

If $y(t)$ is another solution with initial value y_0, then

$$y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds,$$

and hence

$$y(t) - x(t) = y_0 - x_0 + \int_{t_0}^{t} \big(f(s, y(s)) - f(s, x(s))\big)\, ds.$$

Suppose f is Lipschitz continuous in x with Lipschitz constant L, so that

$$\|f(t, y) - f(t, x)\| \le L\|y - x\|.$$

Taking norms gives

$$\|y(t) - x(t)\| \le \|y_0 - x_0\| + \int_{t_0}^{t} L\|y(s) - x(s)\|\, ds. \tag{4}$$

The following inequality controls this accumulation of error.

Lemma 1 (Grönwall's inequality)**.** *Let $u, v \in C([t_0, t_1], \mathbb{R})$ be nonnegative functions and let $\rho \ge 0$. If*

$$u(t) \le \rho + \int_{t_0}^{t} v(s)u(s)\, ds, \qquad t \in [t_0, t_1], \tag{5}$$

then

$$u(t) \le \rho \exp\left(\int_{t_0}^{t} v(s)\, ds\right), \qquad t \in [t_0, t_1]. \tag{6}$$

In particular, $u \equiv 0$ when $\rho = 0$.

Proof. First suppose that $\rho > 0$ and define

$$U(t) = \rho + \int_{t_0}^{t} v(s)u(s)\,ds.$$

Then $U \in C^1([t_0, t_1], \mathbb{R})$, and

$$u(t) \le U(t), \qquad U(t) \ge \rho > 0.$$

Since u and v are nonnegative,

$$U'(t) = v(t)u(t) \le v(t)U(t).$$

Thus the logarithmic derivative of U satisfies

$$\bigl(\log U(t)\bigr)' = \frac{U'(t)}{U(t)} \le v(t).$$

Integrating from t_0 to t yields

$$\log U(t) - \log U(t_0) \le \int_{t_0}^{t} v(s)\,ds.$$

Because $U(t_0) = \rho$ and $u(t) \le U(t)$, applying the exponential function proves (6) when $\rho > 0$.

Now suppose that $\rho = 0$. For any $\varepsilon > 0$, (5) implies

$$u(t) \le \varepsilon + \int_{t_0}^{t} v(s)u(s)\,ds.$$

The first part of the proof, with $\rho = \varepsilon$, gives

$$0 \le u(t) \le \varepsilon \exp\left(\int_{t_0}^{t} v(s)\,ds\right).$$

Letting $\varepsilon \downarrow 0$ shows that $u(t) = 0$ for every $t \in [t_0, t_1]$. □

Apply Lemma 1 to (4) with

$$u(t) = \|y(t) - x(t)\|, \qquad v(t) = L, \qquad \rho = \|y_0 - x_0\|.$$

We obtain

$$\|y(t) - x(t)\| \le \|y_0 - x_0\| e^{L(t-t_0)}. \tag{7}$$

This estimate can be pessimistic, but it establishes continuous dependence on the initial data.

2.2. **The solution map.** The *solution map*, or *flow map*, is denoted by

$$x(t) = \Phi^{t,t_0}(x_0).$$

For example, consider the scalar logistic equation

$$x' = x(1 - x).$$

For $x_0 \neq 0$, its solution is

$$\Phi^{t,t_0}(x_0) = \frac{1}{1 + \dfrac{1 - x_0}{x_0} e^{-(t-t_0)}}.$$

When $x_0 = 0$, the solution is $x(t) \equiv 0$.

The solution map has the identity and composition properties

$$\begin{aligned} \Phi^{t_0,t_0}(x_0) &= x_0, & \Phi^{t_0,t_0} &= I, \\ \Phi^{t_2,t_1}\big(\Phi^{t_1,t_0}(x_0)\big) &= \Phi^{t_2,t_0}(x_0). \end{aligned}$$

The second property is the basis of time marching.

Differentiating the solution map with respect to time gives

$$\begin{aligned} \frac{d}{dt}\Phi^{t,t_0}(x_0) &= f(t, x(t)), \\ \frac{d^2}{dt^2}\Phi^{t,t_0}(x_0) &= f_t(t, x(t)) + f_x(t, x(t))x'(t) \\ &= \big(f_t + f_x f\big)(t, x(t)), \end{aligned}$$

where f_x is the Jacobian matrix of f with respect to x. Higher time derivatives can be found by repeated differentiation, for example with automatic differentiation; they can also be viewed through the Koopman operator.

To measure sensitivity with respect to the initial value, define the Wronskian matrix

$$W(t) = \frac{\partial}{\partial x_0}\Phi^{t,t_0}(x_0) = \frac{\partial x(t)}{\partial x_0}.$$

Differentiating with respect to t gives

$$W'(t) = \frac{\partial}{\partial x_0} f(t, x(t))$$

$$= f_x(t, x(t)) \frac{\partial x(t)}{\partial x_0}.$$

Thus, if

$$A(t) = f_x(t, x(t)),$$

then W satisfies the *variational equation*

$$W'(t) = A(t)W(t), \qquad W(t_0) = I. \tag{8}$$

2.3. **One-step methods.** Choose a time grid

$$t_0 < t_1 < \cdots < t_N = t, \qquad \tau_j = t_{j+1} - t_j,$$

where the step sizes τ_j are small. Repeated use of the composition property gives

$$\Phi^{t_N,t_0} = \Phi^{t_N,t_{N-1}} \circ \cdots \circ \Phi^{t_2,t_1} \circ \Phi^{t_1,t_0}.$$

It is therefore enough to approximate a single short-time solution map. Writing $t_j = t$ and $t_{j+1} = t + \tau$, Taylor expansion gives

$$\Phi^{t+\tau,t}(x) = x + \tau f(t, x) + \frac{\tau^2}{2}\big(f_t + f_x f\big)(t, x) + O(\tau^3). \tag{9}$$

Dropping the terms of order τ^2 and higher defines the Euler one-step map

$$\Psi^{t+\tau,t}(x) = x + \tau f(t, x).$$

Consequently, Euler's method is

$$x_{j+1} = x_j + \tau_j f(t_j, x_j). \tag{10}$$

Its one-step consistency error follows from (9):

$$\Phi^{t+\tau,t}(x) - \Psi^{t+\tau,t}(x) = \frac{\tau^2}{2}\big(f_t + f_x f\big)(t, x) + O(\tau^3) = O(\tau^2).$$

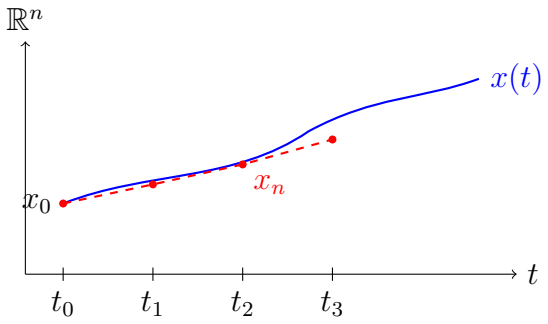
3. August 28, 2026

We continue the study of one-step methods, focusing on consistency, examples, and convergence. Let

$$t_0 < t_1 < \cdots < t_N, \qquad \tau_n = t_{n+1} - t_n,$$

where t_n is a time level and τ_n is the corresponding step size. We use $x(t_n)$ for the exact solution and x_n for its numerical approximation:

$$x(t_n) \approx x_n.$$



A one-step method advances the numerical solution by a map

$$x_{n+1} = \Psi^{t_{n+1},t_n}(x_n). \tag{11}$$

Thus the exact and numerical solutions at t_n can be written as

$$x(t_n) = \Phi^{t_n,t_{n-1}} \circ \cdots \circ \Phi^{t_2,t_1} \circ \Phi^{t_1,t_0}(x_0),$$
$$x_n = \Psi^{t_n,t_{n-1}} \circ \cdots \circ \Psi^{t_2,t_1} \circ \Psi^{t_1,t_0}(x_0).$$

3.1. **Methods obtained from quadrature.** The integral form of the ODE over one step is

$$x(t_{n+1}) = x(t_n) + \int_{t_n}^{t_{n+1}} f(s, x(s))\, ds. \tag{12}$$

Different quadrature approximations of this integral produce different one-step methods.

3.1.1. *Forward Euler method.* Using the left endpoint in (12) gives

$$x(t_{n+1}) \approx x(t_n) + \tau_n f(t_n, x(t_n)).$$

Replacing the exact solution values by numerical approximations yields

$$x_{n+1} = x_n + \tau_n f(t_n, x_n) = \Psi_{\mathrm{FE}}^{t_{n+1},t_n}(x_n). \tag{13}$$

3.1.2. *Backward Euler method.* Using the right endpoint instead gives

$$x(t_{n+1}) \approx x(t_n) + \tau_n f(t_{n+1}, x(t_{n+1})).$$

The resulting method is

$$x_{n+1} = x_n + \tau_n f(t_{n+1}, x_{n+1}) = \Psi_{\mathrm{BE}}^{t_{n+1},t_n}(x_n). \tag{14}$$

Because x_{n+1} occurs on both sides, backward Euler requires the solution of a fixed-point problem at each step.

3.1.3. *A predictor–corrector method.* The trapezoidal rule suggests

$$x(t_{n+1}) \approx x(t_n) + \frac{\tau_n}{2}\big(f(t_n, x(t_n)) + f(t_{n+1}, x(t_{n+1}))\big).$$

We can first predict the right endpoint with forward Euler and then use that prediction in the trapezoidal formula:

$$x_{n+1}^* = x_n + \tau_n f(t_n, x_n), \tag{15}$$

$$x_{n+1} = x_n + \frac{\tau_n}{2}\big(f(t_n, x_n) + f(t_{n+1}, x_{n+1}^*)\big). \tag{16}$$

This explicit two-stage scheme is also called Heun's method. Together, the two stages define a one-step map

$$x_{n+1} = \Psi_{\mathrm{H}}^{t_{n+1},t_n}(x_n).$$

3.2. **Local consistency error.** For a numerical one-step map Ψ, define its *local error*, or *consistency error*, at (t_n, x) by

$$\mathcal{E}_n(x) = \Phi^{t_{n+1},t_n}(x) - \Psi^{t_{n+1},t_n}(x). \tag{17}$$

Here the exact and numerical steps start from the same value x at time t_n.

Example 5 (Forward Euler)**.** *The Taylor expansion from* (9) *gives*

$$\begin{aligned}
\mathcal{E}_n^{\mathrm{FE}}(x) &= \Phi^{t_n+\tau_n,t_n}(x) - \big(x + \tau_n f(t_n, x)\big) \\
&= \frac{\tau_n^2}{2}\big(f_t + f_x f\big)(t_n, x) + O(\tau_n^3) \\
&= O(\tau_n^2).
\end{aligned}$$

Example 6 (Backward Euler)**.** *Let*

$$y(\tau) = \Psi_{\mathrm{BE}}^{t_n+\tau,t_n}(x).$$

The defining fixed-point equation is

$$y(\tau) = x + \tau f(t_n + \tau, y(\tau)).$$

Differentiating this equation and evaluating at $\tau = 0$ *gives*

$$y'(0) = f(t_n, x), \qquad y''(0) = 2\big(f_t + f_x f\big)(t_n, x).$$

Therefore,

$$\Psi_{\mathrm{BE}}^{t_n+\tau_n, t_n}(x) = x + \tau_n f(t_n, x) + \tau_n^2\big(f_t + f_x f\big)(t_n, x) + O(\tau_n^3).$$

Comparing this with the exact flow expansion yields

$$\begin{aligned}\mathcal{E}_n^{\mathrm{BE}}(x) &= -\frac{\tau_n^2}{2}\big(f_t + f_x f\big)(t_n, x) + O(\tau_n^3) \\ &= O(\tau_n^2).\end{aligned}$$

3.3. **Global error and convergence.** The *global error* at t_n is

$$e_n = x_n - x(t_n). \tag{18}$$

The local errors accumulate as the method marches forward in time. On a fixed time interval there are approximately $1/\tau$ steps of size τ, so one generally loses one power of τ when passing from local to global error:

$$\mathcal{E}_n = O(\tau^{p+1}) \quad \Longrightarrow \quad e_n = O(\tau^p).$$

More generally, with

$$h = \max_{0 \le n < N} \tau_n,$$

a stable one-step method with local error $O(h^{p+1})$ has global error $O(h^p)$ under the usual smoothness assumptions. Thus both forward and backward Euler have local error of order two and global order one.

4. August 31, 2026

We continue the analysis of one-step methods. The main goals are to pass from local consistency to global convergence and to introduce Runge–Kutta methods. Consider the IVP

$$x'(t) = f(t, x(t)), \qquad x(t_0) = x_0 \in \mathbb{R}^n,$$

on a grid

$$t_0 < t_1 < \cdots < t_N = T, \qquad \tau_j = t_{j+1} - t_j, \qquad h = \max_{0 \le j < N} \tau_j.$$

4.1. **From local consistency to global convergence.** Recall that Φ^{t_{j+1},t_j} denotes the exact flow over one step, while Ψ^{t_{j+1},t_j} denotes the numerical one-step map. For a starting value x, define the local consistency error by

$$\xi_j(x) = \Phi^{t_{j+1},t_j}(x) - \Psi^{t_{j+1},t_j}(x). \tag{19}$$

Definition 3 (Consistency order)**.** *A one-step method has* consistency order $p > 0$ *if, uniformly for the relevant starting values,*

$$\|\xi_j(x)\| \le C\tau_j^{p+1}$$

as $h \to 0$. The order p need not be an integer when the data have only Hölder regularity.

Forward and backward Euler have consistency order one. The exact and numerical values at the final time can be written as compositions:

$$x(T) = \Phi^{t_N,t_{N-1}} \circ \cdots \circ \Phi^{t_2,t_1} \circ \Phi^{t_1,t_0}(x_0),$$
$$x_N = \Psi^{t_N,t_{N-1}} \circ \cdots \circ \Psi^{t_2,t_1} \circ \Psi^{t_1,t_0}(x_0).$$

Thus the local errors are propagated through all the subsequent steps.

Let

$$e_j = x_j - x(t_j)$$

be the global error. For brevity, set $\Psi_j = \Psi^{t_{j+1},t_j}$ and $\Phi_j = \Phi^{t_{j+1},t_j}$. Then

$$\begin{aligned} e_{j+1} &= \Psi_j(x_j) - \Phi_j(x(t_j)) \\ &= \Psi_j(x_j) - \Psi_j(x(t_j)) - \xi_j(x(t_j)). \end{aligned} \tag{20}$$

Suppose the one-step map satisfies the stability estimate

$$\|\Psi_j(y) - \Psi_j(z)\| \le e^{L\tau_j}\|y - z\|. \tag{21}$$

Combining the consistency and stability estimates in (20) gives

$$\|e_{j+1}\| \le e^{L\tau_j}\|e_j\| + C\tau_j^{p+1}. \tag{22}$$

Since $e_0 = 0$, iteration yields

$$\begin{aligned}\|e_N\| &\le C \sum_{j=0}^{N-1} \tau_j^{p+1} e^{L(T-t_{j+1})} \\ &\le Ch^p \sum_{j=0}^{N-1} \tau_j e^{L(T-t_{j+1})}.\end{aligned}$$

For $L > 0$, the sum is bounded by a geometric integral, and hence

$$\|e_N\| \le \frac{C}{L}\big(e^{L(T-t_0)} - 1\big)h^p. \tag{23}$$

When $L = 0$, the corresponding bound is

$$\|e_N\| \le C(T - t_0)h^p.$$

In either case, $h \to 0$ implies $e_N \to 0$. Thus stability converts a local error of order $p + 1$ into a global error of order p.

Remark 1 (Contractive problems). In the favorable case

$$\|\Psi_j(y) - \Psi_j(z)\| \le \|y - z\|,$$

no exponential amplification occurs. Direct summation gives

$$\|e_N\| \le C \sum_{j=0}^{N-1} \tau_j^{p+1} \le C(T - t_0)h^p.$$

Such contractivity is especially relevant for dissipative ODEs and numerical methods that preserve their dissipative structure.

4.2. **Runge–Kutta methods.** Consider an explicit method with s stages and coefficients a_{ij}, b_i, and c_i. Starting from x at time t with step size τ, its stages are computed successively from

$$k_i = f\left(t + c_i\tau, x + \tau \sum_{j=1}^{i-1} a_{ij}k_j\right), \qquad i = 1, \dots, s, \tag{24}$$

and its one-step map is

$$\Psi^{t+\tau,t}(x) = x + \tau \sum_{i=1}^{s} b_i k_i. \tag{25}$$

Example 7 (Forward Euler)**.** *Take $s = 1$, $c_1 = 0$, $a_{11} = 0$, and $b_1 = 1$. Then*

$$k_1 = f(t, x), \qquad \Psi^{t+\tau,t}(x) = x + \tau k_1,$$

which is forward Euler, a first-order method.

Example 8 (Explicit trapezoidal method)**.** *Take $s = 2$ and*

$$A = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \qquad b = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \end{bmatrix}, \qquad c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

The stages and update are

$$\begin{aligned} k_1 &= f(t, x), \\ k_2 &= f(t + \tau, x + \tau k_1), \\ \Psi^{t+\tau,t}(x) &= x + \frac{\tau}{2}(k_1 + k_2). \end{aligned}$$

This is the predictor–corrector method from the previous lecture, also called Heun's method. It has order two.

5. September 2, 2026

We develop the coefficient representation of Runge–Kutta methods and derive their order conditions. Throughout, we retain the notation of the previous lectures: $\tau_n = t_{n+1} - t_n$, x_n is the numerical approximation at t_n, Φ is the exact flow, and Ψ is the numerical one-step map.

5.1. Butcher tableaux.

An s-stage Runge–Kutta method has the form

$$k_i = f\left(t_n + c_i\tau_n, x_n + \tau_n \sum_{j=1}^{s} a_{ij} k_j\right), \qquad i = 1, \dots, s, \tag{26}$$

$$x_{n+1} = x_n + \tau_n \sum_{i=1}^{s} b_i k_i. \tag{27}$$

The coefficients are collected into

$$A = (a_{ij})_{i,j=1}^{s}, \qquad b = \begin{bmatrix} b_1 & \cdots & b_s \end{bmatrix}^{\mathsf{T}}, \qquad c = \begin{bmatrix} c_1 & \cdots & c_s \end{bmatrix}^{\mathsf{T}}.$$

They are conventionally displayed in a *Butcher tableau*:

$$\begin{array}{c|cccc} c_1 & a_{11} & a_{12} & \cdots & a_{1s} \\ c_2 & a_{21} & a_{22} & \cdots & a_{2s} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ c_s & a_{s1} & a_{s2} & \cdots & a_{ss} \\ \hline & b_1 & b_2 & \cdots & b_s \end{array}.$$

The values $t_n + c_i\tau_n$ are the intermediate stage times between t_n and t_{n+1}. If A is strictly lower triangular, then each stage depends only on previously computed stages. The resulting method is an *explicit Runge–Kutta method*.

5.2. **The internal consistency relation.** Any nonautonomous ODE can be converted into an autonomous system by augmenting its state. Set

$$z(t) = \begin{bmatrix} t \\ x(t) \end{bmatrix}, \qquad z'(t) = \begin{bmatrix} 1 \\ f(t, x(t)) \end{bmatrix} = F(z(t)).$$

Apply a Runge–Kutta method to this autonomous system. Because the time component of every stage slope is one, the time component of stage i is

$$t_n + \tau_n \sum_{j=1}^{s} a_{ij}.$$

For this to agree with the prescribed stage time $t_n + c_i\tau_n$ in (26), we require

$$c_i = \sum_{j=1}^{s} a_{ij}, \qquad i = 1, \ldots, s.$$

Thus the *internal consistency relation* is

$$c = A\mathbf{1}, \qquad \mathbf{1} = \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix}^{\mathsf{T}}. \tag{28}$$

Under this relation, applying the Runge–Kutta scheme directly to the nonautonomous equation is equivalent to applying it to the augmented autonomous system.

5.3. **One-step error and order.** For an exact starting value x at time t, define the one-step defect by

$$\mathcal{E}(x,t,\tau) = \Phi^{t+\tau,t}(x) - \Psi^{t+\tau,t}(x). \tag{29}$$

A Runge–Kutta method has order p if

$$\mathcal{E}(x,t,\tau) = O(\tau^{p+1}) \qquad \text{as } \tau \to 0.$$

Assuming (28), it suffices to derive the conditions for an autonomous equation

$$x' = f(x).$$

We write its exact and Runge–Kutta flows as Φ^τ and Ψ^τ, respectively.

Let $f' = Df(x)$ and $f'' = D^2 f(x)$. Differentiation along an exact solution gives

$$\begin{aligned} x' &= f, \\ x'' &= f'f, \\ x''' &= f''(f,f) + f'f'f. \end{aligned}$$

Consequently,

$$\Phi^\tau(x) = x + \tau f + \frac{\tau^2}{2} f'f + \frac{\tau^3}{6}\big(f''(f,f) + f'f'f\big) + O(\tau^4). \tag{30}$$

For the numerical method, set

$$\Delta_i = \tau \sum_{j=1}^{s} a_{ij} k_j, \qquad k_i = f(x + \Delta_i).$$

Since $k_j = f + O(\tau)$ and $c = A\mathbf{1}$,

$$\Delta_i = \tau c_i f + O(\tau^2).$$

A more detailed Taylor expansion gives

$$k_i = f + \tau c_i f'f + \tau^2 \left(\sum_{j=1}^{s} a_{ij} c_j \, f'f'f + \frac{c_i^2}{2} f''(f,f) \right) + O(\tau^3).$$

Substitution into $\Psi^\tau(x) = x + \tau \sum_i b_i k_i$ yields

$$\Psi^\tau(x) = x + \tau(b^{\mathsf{T}}\mathbf{1}) f + \tau^2 (b^{\mathsf{T}} c) f'f$$

$$+\tau^3\left((b^{\mathsf{T}}Ac)f'f'f+\frac{1}{2}(b^{\mathsf{T}}c^{\circ 2})f''(f,f)\right)+O(\tau^4), \tag{31}$$

where $c^{\circ 2}=(c_1^2,\dots,c_s^2)^{\mathsf{T}}$.

Comparing (30) and (31) gives the order conditions through order three:

$$b^{\mathsf{T}}\mathbf{1}=1, \qquad \text{(order one)}, \tag{32}$$

$$b^{\mathsf{T}}c=\frac{1}{2}, \qquad \text{(additional condition for order two)}, \tag{33}$$

$$b^{\mathsf{T}}c^{\circ 2}=\frac{1}{3}, \qquad \text{(additional condition for order three)}, \tag{34}$$

$$b^{\mathsf{T}}Ac=\frac{1}{6}, \qquad \text{(additional condition for order three)}. \tag{35}$$

Therefore (32) gives a first-order method; the first two conditions give a second-order method; and all four give a third-order method. A fourth-order method must satisfy eight conditions in total. For nonautonomous equations, the relation $c=A\mathbf{1}$ is also needed for these autonomous order conditions to apply directly.

5.4. **Examples.**

Example 9 (Forward Euler)**.** *Forward Euler has the tableau*

$$\begin{array}{c|c} 0 & 0 \\ \hline & 1 \end{array}.$$

Thus $s=1$, $A=0$, $b=1$, and $c=0$. It satisfies $b^{\mathsf{T}}\mathbf{1}=1$, but not the second-order condition, so it is first order.

Example 10 (Explicit trapezoidal method)**.** *The explicit trapezoidal method, or Heun's method, is*

$$\begin{aligned} k_1 &= f(t_n,x_n),\\ k_2 &= f(t_n+\tau_n,x_n+\tau_n k_1),\\ x_{n+1} &= x_n+\frac{\tau_n}{2}(k_1+k_2). \end{aligned}$$

Its tableau is

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & 1 & 0 \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}.$$

Here $c = A\mathbf{1}$, $b^{\mathsf{T}}\mathbf{1} = 1$, *and* $b^{\mathsf{T}}c = 1/2$. *However,* $b^{\mathsf{T}}c^{\circ 2} = 1/2 \neq 1/3$, *so the method is second order but not third order.*

5.5. **Explicit three-stage methods.** A general explicit three-stage Runge–Kutta method has tableau

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ c_2 & a_{21} & 0 & 0 \\ c_3 & a_{31} & a_{32} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}.$$

Internal consistency requires

$$c_2 = a_{21}, \qquad c_3 = a_{31} + a_{32}.$$

Before imposing the order conditions, the six free coefficients are

$$b_1, b_2, b_3, a_{21}, a_{31}, a_{32}.$$

The four conditions (32)–(35) enforce third order, so one generically expects a two-parameter family of explicit three-stage schemes of order three.

6. September 4, 2026

We continue the study of Runge–Kutta methods, first constructing families of explicit methods and then introducing embedded methods and stability functions. We use the notation established in the previous lectures: $\tau_n = t_{n+1} - t_n$, x_n is the numerical solution, and Φ and Ψ denote the exact and numerical flows.

6.1. **Runge–Kutta methods and order conditions.** Recall that an s-stage Runge–Kutta method is

$$k_i = f\left(t_n + c_i\tau_n, x_n + \tau_n \sum_{j=1}^{s} a_{ij}k_j\right), \qquad i = 1, \dots, s,$$

$$x_{n+1} = x_n + \tau_n \sum_{i=1}^{s} b_i k_i.$$

Here s is the number of stages, $A = (a_{ij})$ contains the stage coefficients, c_i is the i-th stage node, and b_i is the i-th quadrature weight. The internal consistency relation is

$$c = A\mathbf{1}.$$

It makes the method agree on the original nonautonomous equation and its autonomous augmentation. The method is explicit precisely when A is strictly lower triangular.

For the autonomous equation $x' = f(x)$, a method has order p when

$$\left\|\Phi^\tau(x) - \Psi^\tau(x)\right\| = O(\tau^{p+1}) \qquad \text{as } \tau \to 0.$$

The first four order conditions, derived in the previous lecture, are

$$b^\mathsf{T}\mathbf{1} = 1, \qquad b^\mathsf{T}c = \frac{1}{2},$$
$$b^\mathsf{T}c^{\circ 2} = \frac{1}{3}, \qquad b^\mathsf{T}Ac = \frac{1}{6}.$$

They are sufficient through order three. To obtain order four, four more conditions are needed:

$$b^\mathsf{T}c^{\circ 3} = \frac{1}{4}, \qquad b^\mathsf{T}\big(c \circ (Ac)\big) = \frac{1}{8},$$
$$b^\mathsf{T}A(c^{\circ 2}) = \frac{1}{12}, \qquad b^\mathsf{T}A^2c = \frac{1}{24},$$

where $\circ$ denotes componentwise multiplication. Thus one, two, four, and eight conditions must be satisfied for orders one, two, three, and four, respectively.

Forward Euler uses one stage and has order one, while the explicit trapezoidal method uses two stages and has order two. We next continue the construction of higher-order explicit methods.

6.2. **Explicit three-stage methods of order three.** For an explicit three-stage method, internal consistency gives the tableau

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ c_2 & a_{21} & 0 & 0 \\ c_3 & a_{31} & a_{32} & 0 \\ \hline & b_1 & b_2 & b_3 \end{array}, \qquad c_2 = a_{21}, \quad c_3 = a_{31} + a_{32}.$$

There are six free coefficients and four order conditions, so the third-order methods generically form a two-parameter family. Let

$$c_2 = \alpha, \qquad c_3 = \beta.$$

Solving the four order conditions gives

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \alpha & \alpha & 0 & 0 \\ \beta & \dfrac{\beta\big(3\alpha(1-\alpha)-\beta\big)}{\alpha(2-3\alpha)} & \dfrac{\beta(\beta-\alpha)}{\alpha(2-3\alpha)} & 0 \\ \hline & \dfrac{6\alpha\beta-3\alpha-3\beta+2}{6\alpha\beta} & \dfrac{3\beta-2}{6\alpha(\beta-\alpha)} & \dfrac{2-3\alpha}{6\beta(\beta-\alpha)} \end{array},$$

provided the displayed denominators are nonzero.

A useful subfamily has $\beta = 1$. Its weights are

$$b_1 = \frac{3\alpha - 1}{6\alpha}, \qquad b_2 = \frac{1}{6\alpha(1-\alpha)}, \qquad b_3 = \frac{2-3\alpha}{6(1-\alpha)}.$$

All three weights are nonnegative when $\alpha \in [1/3, 2/3)$, but positivity of the weights alone does not guarantee that one method is better than another.

Example 11 (Kutta's third-order method)**.** *Taking $\alpha = 1/2$ and $\beta = 1$ gives*

$$\begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 \\ 1 & -1 & 2 & 0 \\ \hline & \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{array}.$$

Thus

$$k_1 = f(t_n, x_n),$$

$$
\begin{aligned}
k_2 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_1\right), \\
k_3 &= f\left(t_n + \tau_n, x_n + \tau_n(-k_1 + 2k_2)\right), \\
x_{n+1} &= x_n + \frac{\tau_n}{6}(k_1 + 4k_2 + k_3).
\end{aligned}
$$

6.3. **Explicit four-stage methods of order four.** Before imposing internal consistency, a general explicit four-stage tableau contains thirteen coefficients: three nontrivial nodes, six strictly lower-triangular entries of A, and four weights. Internal consistency removes three degrees of freedom, and the eight fourth-order conditions generically leave two free parameters.

In the usual nonconfluent family one may take

$$
c_1 = 0, \qquad c_2 = \alpha, \qquad c_3 = \beta, \qquad c_4 = 1,
$$

with distinct stage nodes. The generic formulas exclude degenerate parameter choices; one convenient nondegeneracy condition is

$$
\alpha\beta(\alpha-1)(\beta-1)(\alpha-\beta)(2\alpha-1)\big(6\alpha\beta - 4\alpha - 4\beta + 3\big) \neq 0.
$$

Example 12 (The three-eighths rule)**.** *Choosing $\alpha = 1/3$ and $\beta = 2/3$ produces the fourth-order three-eighths rule*

$$
\begin{array}{c|cccc}
0 & 0 & 0 & 0 & 0 \\
\frac{1}{3} & \frac{1}{3} & 0 & 0 & 0 \\
\frac{2}{3} & -\frac{1}{3} & 1 & 0 & 0 \\
1 & 1 & -1 & 1 & 0 \\
\hline
 & \frac{1}{8} & \frac{3}{8} & \frac{3}{8} & \frac{1}{8}
\end{array}.
$$

Its stages are

$$
\begin{aligned}
k_1 &= f(t_n, x_n), \\
k_2 &= f\left(t_n + \frac{\tau_n}{3}, x_n + \frac{\tau_n}{3}k_1\right), \\
k_3 &= f\left(t_n + \frac{2\tau_n}{3}, x_n + \tau_n\left(-\frac{1}{3}k_1 + k_2\right)\right), \\
k_4 &= f\left(t_n + \tau_n, x_n + \tau_n(k_1 - k_2 + k_3)\right), \\
x_{n+1} &= x_n + \frac{\tau_n}{8}(k_1 + 3k_2 + 3k_3 + k_4).
\end{aligned}
$$

Repeated nodes give a separate, *confluent* family. Setting $c_2 = c_3 = 1/2$ and letting $\delta = a_{32} \neq 0$ yields

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2}-\delta & \delta & 0 & 0 \\ 1 & 0 & 1-\dfrac{1}{2\delta} & \dfrac{1}{2\delta} & 0 \\ \hline & \frac{1}{6} & \dfrac{4\delta-1}{6\delta} & \dfrac{1}{6\delta} & \frac{1}{6} \end{array}. \tag{36}$$

Example 13 (Classical RK4)**.** *The choice $\delta = 1/2$ in (36) gives the classical tableau*

$$\begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 & 0 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}.$$

Equivalently,

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_1\right), \\ k_3 &= f\left(t_n + \frac{\tau_n}{2}, x_n + \frac{\tau_n}{2}k_2\right), \\ k_4 &= f(t_n + \tau_n, x_n + \tau_n k_3), \\ x_{n+1} &= x_n + \frac{\tau_n}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

6.4. **Embedded pairs.** Adaptive solvers need local error estimates. They obtain them by combining the same stages in two different ways. Given fixed A and c, choose two weight vectors b and $\widehat{b}$ and form

$$x_{n+1}^{[p]} = x_n + \tau_n \sum_{i=1}^{s} b_i k_i,$$

$$x_{n+1}^{[p+1]} = x_n + \tau_n \sum_{i=1}^{s} \widehat{b}_i k_i.$$

The two formulas share all evaluations k_i, so

$$\mathrm{err}_n = \left\| x_{n+1}^{[p+1]} - x_{n+1}^{[p]} \right\|$$

is an inexpensive local error estimate. If this quantity is well below the requested tolerance, the next step size can be increased. If it is too large, the step size is decreased and the step may be rejected.

MATLAB's `ode45` uses a seven-stage Dormand–Prince embedded pair of orders four and five. The minimal number of stages for an explicit Runge–Kutta method grows with its order; the low-order values discussed in class are

p	1	2	3	4	5	6	7	8
$s_{\min}$	1	2	3	4	6	7	9	11

6.5. **The stability function.** Apply a Runge–Kutta method to the linear test equation

$$y'(t) = \lambda y(t). \tag{37}$$

With $z = \tau_n \lambda$, one numerical step has the form

$$y_{n+1} = R(z) y_n, \tag{38}$$

where R is the *stability function*. The exact amplification factor is e^z.

For a Runge–Kutta method, the stage equations applied to (37) give

$$(I - zA)k = \lambda y_n \mathbf{1}.$$

Substituting the resulting stages into the update gives

$$R(z) = 1 + z b^{\mathsf{T}} (I - zA)^{-1} \mathbf{1}. \tag{39}$$

If the method has order p, then necessarily

$$R(z) - e^z = O(z^{p+1}) \qquad \text{as } z \to 0. \tag{40}$$

For example, the eight fourth-order conditions imply

$$R(z) = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \frac{z^4}{4!} + O(z^5).$$

Condition (40) is necessary for order p, but it only tests a linear ODE and is not sufficient to establish order p for general nonlinear equations.

7. September 9, 2026

This lecture introduces extrapolation as a way to increase the order of an existing time-stepping method. We then begin the study of stability as a property of solutions of the differential equation itself.

7.1. **Accuracy and computational cost.**

Consider a method of order p on the uniform grid

$$t_n = t_0 + n\tau, \qquad n = 0, \dots, N, \qquad t_N = T,$$

and let $Y_\tau(T) = x_N$ denote the numerical approximation at the final time. Convergence of order p means that, for sufficiently small τ,

$$\big\|Y_\tau(T) - x(T)\big\| \le C(T)\tau^p. \tag{41}$$

In particular, $Y_\tau(T) \to x(T)$ as $\tau \to 0$. For a fixed step size, increasing the order generally decreases the error, provided that τ is in the asymptotic regime in which (41) is valid.

Suppose a tolerance $\varepsilon > 0$ is prescribed. Balancing the error estimate with the tolerance gives

$$C(T)\tau^p \approx \varepsilon, \qquad \tau \approx \left(\frac{\varepsilon}{C(T)}\right)^{1/p}.$$

Since $N = (T - t_0)/\tau$, the resulting work, measured by the number of time steps, is approximately

$$N \approx (T - t_0)\left(\frac{C(T)}{\varepsilon}\right)^{1/p}. \tag{42}$$

Thus a higher-order method has a more favorable dependence on the tolerance. Formula (42) does not include the work per step or the size of the error constant, both of which also matter in practice.

7.2. **Richardson extrapolation.** Extrapolation requires more than an error bound. Assume the stronger asymptotic expansion

$$Y_\tau(T) = x(T) + c_p(T)\tau^p + c_{p+1}(T)\tau^{p+1} + c_{p+2}(T)\tau^{p+2} + \cdots . \tag{43}$$

Rerunning the same method with half the step size gives

$$Y_{\tau/2}(T) = x(T) + c_p(T)\frac{\tau^p}{2^p} + c_{p+1}(T)\frac{\tau^{p+1}}{2^{p+1}} + O(\tau^{p+2}).$$

Multiplying this equation by 2^p and subtracting (43) cancels the leading error term:

$$2^p Y_{\tau/2}(T) - Y_\tau(T) = (2^p - 1)x(T) + O(\tau^{p+1}).$$

Consequently, the *Richardson extrapolate*

$$Y_\tau^{[1]}(T) = \frac{2^p Y_{\tau/2}(T) - Y_\tau(T)}{2^p - 1} \tag{44}$$

satisfies

$$Y_\tau^{[1]}(T) = x(T) + O(\tau^{p+1}).$$

This gains one order. The underlying one-step method is unchanged; the additional accuracy comes from combining two complete runs.

7.2.1. *Symmetric methods.* For an autonomous problem, a one-step method is *symmetric* if a step of length $-\tau$ exactly reverses a step of length τ:

$$\Psi^{-\tau} = \left(\Psi^\tau\right)^{-1}.$$

Symmetric methods have even order. Under the usual smoothness assumptions, their fixed-time global error expansion contains powers of one parity. If the method has order p, the expansion takes the form

$$Y_\tau(T) = x(T) + d_p(T)\tau^p + d_{p+2}(T)\tau^{p+2} + d_{p+4}(T)\tau^{p+4} + \cdots . \tag{45}$$

Applying the same combination as in (44) cancels the τ^p term, but now there is no τ^{p+1} term. Hence

$$\frac{2^p Y_{\tau/2}(T) - Y_\tau(T)}{2^p - 1} = x(T) + O(\tau^{p+2}),$$

so one extrapolation raises the order by two.

7.3. **Extrapolation from several step sizes.** The preceding cancellation can be repeated using several numerical solutions. Choose distinct positive integers $n_1, \dots, n_m$, set

$$\tau_i = \frac{\bar{\tau}}{n_i}, \qquad i = 1, \dots, m,$$

and compute $Y_{\tau_i}(T)$ for every i. Motivated by (43), fit the generalized polynomial

$$P(h) = a_0 + \sum_{j=0}^{m-2} a_{p+j} h^{p+j} \tag{46}$$

to these values:

$$P(\tau_i) = Y_{\tau_i}(T), \qquad i = 1, \dots, m. \tag{47}$$

The unknowns $a_0, a_p, \dots, a_{p+m-2}$ are determined by the linear system

$$\begin{bmatrix} 1 & \tau_1^p & \tau_1^{p+1} & \cdots & \tau_1^{p+m-2} \\ 1 & \tau_2^p & \tau_2^{p+1} & \cdots & \tau_2^{p+m-2} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \tau_m^p & \tau_m^{p+1} & \cdots & \tau_m^{p+m-2} \end{bmatrix} \begin{bmatrix} a_0 \\ a_p \\ a_{p+1} \\ \vdots \\ a_{p+m-2} \end{bmatrix} = \begin{bmatrix} Y_{\tau_1}(T) \\ Y_{\tau_2}(T) \\ \vdots \\ Y_{\tau_m}(T) \end{bmatrix}.$$

For a vector-valued ODE, the same coefficient matrix is used for every component. Extrapolating to zero step size gives

$$\overline{x}(T) = P(0) = a_0. \tag{48}$$

The first $m-1$ error terms have been removed, so, under the assumed full expansion,

$$\overline{x}(T) = x(T) + O(\bar{\tau}^{p+m-1}).$$

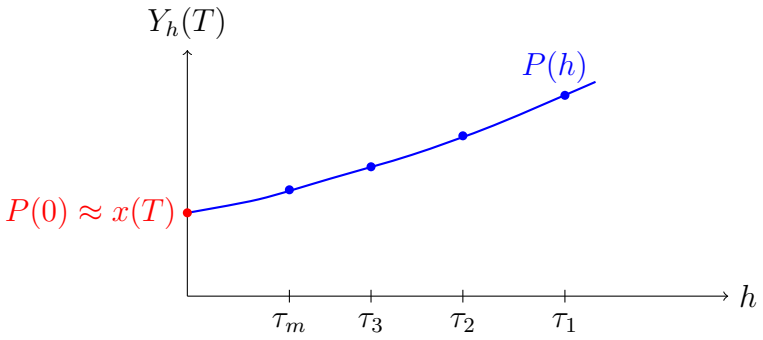
Equivalently, a_0 is a linear combination of the computed values,

$$a_0 = \sum_{i=1}^{m} \gamma_i Y_{\tau_i}(T),$$

where the weights are chosen so that

$$\sum_{i=1}^{m} \gamma_i = 1, \qquad \sum_{i=1}^{m} \gamma_i \tau_i^{p+j} = 0, \quad j = 0, \dots, m-2.$$

For $m = 2$, $n_1 = 1$, and $n_2 = 2$, these conditions recover (44). For a symmetric error expansion, one instead uses the powers $p, p+2, p+4, \ldots$ in (46).



7.4. **Stability of solutions.** Return to the IVP

$$x'(t) = f(t, x(t)), \qquad x(t_0) = x_0, \tag{49}$$

and write its solution as $x(t; t_0, x_0)$ to display its dependence on the initial data.

Definition 4 (Stability of a solution)**.** *The solution through x_0 is* stable *if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that*

$$\|x_0 - y_0\| < \delta \quad \Longrightarrow \quad \big\|x(t; t_0, x_0) - x(t; t_0, y_0)\big\| < \varepsilon$$

for every $t \geq t_0$.

The estimate from Grönwall's inequality in (7) gives continuous dependence over every fixed finite interval. Stability is stronger: the same δ must control the separation for all future times.

Definition 5 (Asymptotic stability)**.** *A stable solution through x_0 is* asymptotically stable *if all sufficiently close initial values y_0 also satisfy*

$$\big\|x(t; t_0, x_0) - x(t; t_0, y_0)\big\| \longrightarrow 0 \qquad \text{as } t \to \infty.$$

For an autonomous equation $x' = f(x)$, a point x_* satisfying

$$f(x_*) = 0$$

is an *equilibrium*. The solution starting at the equilibrium is constant: if $x_0 = x_*$, then $x(t; t_0, x_*) = x_*$ for all $t \geq t_0$. Stability of this equilibrium is obtained from the preceding definitions by comparing this constant solution with solutions starting at nearby points.

Example 14 (Linear autonomous system)**.** *Consider*

$$x' = Ax.$$

The origin is an equilibrium, and

$$x(t; t_0, x_0) = e^{(t-t_0)A} x_0.$$

Thus the difference of two solutions is

$$x(t; t_0, x_0) - x(t; t_0, y_0) = e^{(t-t_0)A}(x_0 - y_0).$$

The system is stable precisely when e^{tA} remains bounded for $t \geq 0$, and it is asymptotically stable precisely when $e^{tA} \to 0$. In spectral terms, stability requires every eigenvalue to have nonpositive real part and every eigenvalue on the imaginary axis to be semisimple. Asymptotic stability holds exactly when every eigenvalue has strictly negative real part.

In the scalar case $x' = \lambda x$, the solution is $x(t) = e^{\lambda(t-t_0)} x_0$. Hence the equilibrium at the origin is stable when $\operatorname{Re} \lambda \leq 0$, and it is asymptotically stable when $\operatorname{Re} \lambda < 0$.

These notions describe the behavior of the exact differential equation. They are distinct from numerical absolute stability, which asks whether a time-stepping method reproduces the decay of a stable test equation.

8. September 11, 2026

We continue the discussion of stability from the previous lecture. The main goals are to linearize an autonomous ODE near an equilibrium, to study the resulting system using the matrix exponential, and to extend the solution formula to time-dependent linear systems by means of the Dyson series.

8.1. **Linearization near an equilibrium.** Consider the autonomous nonlinear equation

$$x' = f(x) \tag{50}$$

and let x_* be an equilibrium, so that $f(x_*) = 0$. Introduce the perturbation

$$y = x - x_*.$$

Since x_* is constant, $y' = x'$. If f is continuously differentiable, its expansion near x_* is

$$f(x_* + y) = f(x_*) + f_x(x_*)y + r(y) = Ay + r(y), \qquad A = f_x(x_*),$$

where

$$\frac{\|r(y)\|}{\|y\|} \longrightarrow 0 \qquad \text{as } y \to 0.$$

Thus the exact perturbation equation is

$$y' = Ay + r(y), \tag{51}$$

and its linearization is

$$y' = Ay. \tag{52}$$

The equilibrium x_* corresponds to the zero solution of the perturbation equation. Hence studying whether $x(t)$ remains close to x_* begins with studying whether solutions of (52) remain close to zero.

8.2. **Autonomous linear systems and the matrix exponential.** Consider the initial-value problem

$$y' = Ay, \qquad y(t_0) = y_0, \tag{53}$$

where $A \in \mathbb{R}^{n\times n}$ is constant. Repeated differentiation gives

$$y^{(k)}(t_0) = A^k y_0.$$

The Taylor series about t_0 therefore gives

$$y(t) = y_0 + (t - t_0)Ay_0 + \frac{(t - t_0)^2}{2!}A^2 y_0 + \cdots$$

$$= \left(\sum_{k=0}^{\infty} \frac{(t-t_0)^k A^k}{k!} \right) y_0.$$

Definition 6 (Matrix exponential)**.** *For a square matrix B, its exponential is*

$$e^B = \sum_{k=0}^{\infty} \frac{B^k}{k!}. \tag{54}$$

The series converges in every matrix norm. With this notation, the solution of (53) is

$$y(t) = e^{(t-t_0)A} y_0. \tag{55}$$

Indeed,

$$\frac{d}{dt} e^{(t-t_0)A} = A e^{(t-t_0)A} = e^{(t-t_0)A} A, \qquad e^{0A} = I.$$

As observed in the previous lecture, the zero solution of (53) is stable exactly when

$$\sup_{t \geq 0} \|e^{tA}\| < \infty, \tag{56}$$

and it is asymptotically stable exactly when

$$\|e^{tA}\| \longrightarrow 0 \qquad \text{as } t \to \infty. \tag{57}$$

These statements follow directly from

$$y(t; t_0, y_0) - y(t; t_0, z_0) = e^{(t-t_0)A}(y_0 - z_0).$$

Linearity also implies that stability of the zero solution is equivalent to stability of every solution.

8.3. **Time-dependent linear systems and the Dyson series.** Now consider the nonautonomous linear equation

$$y' = A(t)y, \qquad y(t_0) = y_0, \tag{58}$$

where $A(t)$ is continuous. Its integral equation is

$$y(t) = y_0 + \int_{t_0}^{t} A(t_1) y(t_1)\, dt_1. \tag{59}$$

At an intermediate time t_1, the same equation reads

$$y(t_1) = y_0 + \int_{t_0}^{t_1} A(t_2)y(t_2)\,dt_2.$$

Substituting this formula into (59) gives

$$y(t) = y_0 + \left(\int_{t_0}^{t} A(t_1)\,dt_1\right) y_0 + \int_{t_0}^{t}\int_{t_0}^{t_1} A(t_1)A(t_2)y(t_2)\,dt_2\,dt_1.$$

Continuing this substitution produces the *Dyson series*

$$y(t) = U(t,t_0)y_0, \tag{60}$$

where the evolution operator is

$$U(t,t_0) = I + \sum_{k=1}^{\infty}\int_{t_0}^{t}\int_{t_0}^{t_1}\cdots\int_{t_0}^{t_{k-1}} A(t_1)A(t_2)\cdots A(t_k)\,dt_k\cdots dt_2\,dt_1. \tag{61}$$

If $\|A(s)\| \le M$ on $[t_0,t]$, the norm of the k-th term is at most

$$\frac{M^k(t-t_0)^k}{k!},$$

which proves convergence of the series. Termwise differentiation shows that

$$\frac{\partial}{\partial t}U(t,t_0) = A(t)U(t,t_0), \qquad U(t_0,t_0) = I.$$

Because matrices evaluated at different times need not commute, the order $t_1 \ge t_2 \ge \cdots \ge t_k$ in (61) is essential. A common shorthand is the *time-ordered exponential*

$$U(t,t_0) = \mathcal{T}\exp\left(\int_{t_0}^{t} A(s)\,ds\right). \tag{62}$$

If $A(s)A(r) = A(r)A(s)$ for every r,s, time ordering is unnecessary and

$$U(t,t_0) = \exp\left(\int_{t_0}^{t} A(s)\,ds\right).$$

In particular, when $A(t) \equiv A$, the ordered integration region has volume $(t-t_0)^k/k!$, and the Dyson series reduces to $e^{(t-t_0)A}$.

8.4. **Properties of the matrix exponential.** The power-series definition immediately gives several useful identities.

Proposition 1. *Let A and B be square matrices of the same size.*

(i) *If M is invertible, then*

$$e^{MAM^{-1}} = Me^{A}M^{-1}.$$

(ii) *If $AB = BA$, then*

$$e^{A+B} = e^{A}e^{B} = e^{B}e^{A}.$$

(iii) *For a block-diagonal matrix,*

$$e^{\operatorname{diag}(A_1,\dots,A_m)} = \operatorname{diag}(e^{A_1},\dots,e^{A_m}).$$

Proof. For the first identity, observe that $(MAM^{-1})^k = MA^kM^{-1}$ and apply (54). The third identity follows similarly because powers preserve block-diagonal structure. When A and B commute, the binomial theorem is valid for $(A+B)^k$. Substitution into the exponential series and collection of like terms gives the second identity. □

8.5. **Jordan form and linear stability.** The preceding identities reduce the stability analysis of $y' = Ay$ to the analysis of individual Jordan blocks. Over $\mathbb{C}$, write

$$A = PJP^{-1}, \qquad J = \operatorname{diag}(J_1,\dots,J_m). \tag{63}$$

If J_k is a Jordan block of size q_k associated with the eigenvalue λ_k, then

$$J_k = \lambda_k I + N_k, \qquad N_k^{q_k} = 0.$$

Since $\lambda_k I$ commutes with N_k,

$$e^{tJ_k} = e^{t\lambda_k}\left(I + tN_k + \frac{t^2N_k^2}{2!} + \cdots + \frac{t^{q_k-1}N_k^{q_k-1}}{(q_k-1)!}\right). \tag{64}$$

Thus each Jordan block contributes an exponential factor multiplied by a polynomial in t.

Theorem 1 (Stability criterion for a linear system)**.** *For $y' = Ay$, the following statements hold.*

(i) *The zero solution is asymptotically stable if and only if*

$$\operatorname{Re}\lambda < 0 \qquad \text{for every } \lambda \in \sigma(A).$$

(ii) *The zero solution is stable if and only if every eigenvalue satisfies* $\operatorname{Re}\lambda \le 0$ *and every eigenvalue on the imaginary axis is semisimple, meaning that all of its Jordan blocks have size one.*

Proof. By Proposition 1 and (63),

$$e^{tA} = Pe^{tJ}P^{-1} = P\,\operatorname{diag}(e^{tJ_1}, \dots, e^{tJ_m})P^{-1}.$$

If $\operatorname{Re}\lambda_k < 0$, the exponential decay in (64) dominates every polynomial factor, so the block tends to zero. If $\operatorname{Re}\lambda_k > 0$, the block has an exponentially growing solution. When $\operatorname{Re}\lambda_k = 0$, the scalar exponential has constant magnitude. The block is therefore bounded precisely when its nilpotent part is zero, or equivalently when its size is one. Combining the blocks with (56) and (57) proves both assertions. □

Example 15 (A defective eigenvalue on the imaginary axis)**.** *Let*

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The only eigenvalue is zero, but its Jordan block has size two. Since $A^2 = 0$*,*

$$e^{tA} = I + tA = \begin{bmatrix} 1 & t \\ 0 & 1 \end{bmatrix}.$$

Hence

$$y_2(t) = y_2(0), \qquad y_1(t) = y_1(0) + t\,y_2(0).$$

Arbitrarily small initial data with $y_2(0) \neq 0$ *lead to linear growth, so the origin is not stable. This illustrates why nonpositive real parts alone are insufficient for stability.*

For the nonlinear perturbation equation (51), the linearization principle gives two decisive cases. If the spectrum of $f_x(x_*)$ lies in the open left half-plane, then x_* is locally asymptotically stable. If the spectrum contains an eigenvalue with positive real part, then x_* is unstable. If it meets the imaginary axis but not the open right half-plane, linearization alone may be inconclusive; the nonlinear remainder $r(y)$ can determine the behavior.

9. September 14, 2026

We develop two complementary approaches to stability. Linearization transfers spectral information from a Jacobian matrix to a nonlinear equilibrium, while Lyapunov functions establish stability without computing the solution explicitly. We then apply the same ideas to time-dependent linear equations and to the discrete evolution produced by Runge–Kutta methods.

9.1. **From linear to nonlinear stability.** Recall the autonomous equation

$$x' = f(x),$$

an equilibrium x_* satisfying $f(x_*) = 0$, and its Jacobian

$$A = f_x(x_*).$$

With $y = x - x_*$, the perturbation equation derived in (51) is

$$y' = Ay + r(y), \qquad \frac{\|r(y)\|}{\|y\|} \longrightarrow 0 \quad \text{as } y \to 0.$$

The matrix A is called *Hurwitz* if all of its eigenvalues lie in the open left half-plane:

$$\operatorname{Re}\lambda < 0 \qquad \text{for every } \lambda \in \sigma(A).$$

The linearization theorem gives the following nonlinear conclusion.

Theorem 2 (Linearization criterion)**.** *Suppose f is continuously differentiable near x_*.*

(i) *If $f_x(x_*)$ is Hurwitz, then x_* is locally asymptotically stable.*
(ii) *If $f_x(x_*)$ has an eigenvalue with positive real part, then x_* is unstable.*

If no eigenvalue of $f_x(x_*)$ lies on the imaginary axis, the equilibrium is *hyperbolic*. The Hartman–Grobman theorem says that the nonlinear flow near a hyperbolic equilibrium is locally topologically equivalent to its linearization $y' = Ay$. If the spectrum meets the imaginary axis, the linearized system alone may not determine stability.

9.2. **Lyapunov's direct method.** Eigenvalue calculations or explicit solution formulas can be expensive. Lyapunov's method instead measures the size of a solution by a scalar energy-like function.

Definition 7 (Lyapunov function)**.** *A continuously differentiable function V, defined near an equilibrium x_*, is a* Lyapunov function *if*

$$V(x_*) = 0, \qquad V(x) > 0 \quad \text{for } x \neq x_*$$

near x_*, and its derivative along solutions,

$$\dot{V}(x) = \nabla V(x)^{\mathsf{T}} f(x), \tag{65}$$

is nonpositive.

Theorem 3 (Lyapunov stability theorem)**.** *If V is positive definite near x_* and $\dot{V} \leq 0$, then x_* is stable. If, in addition, $\dot{V}(x) < 0$ for every $x \neq x_*$ sufficiently close to x_*, then x_* is locally asymptotically stable.*

The idea is that a solution cannot cross outward through a sufficiently small level set of V when $V(x(t))$ is nonincreasing. Strict decrease forces the solution toward the equilibrium.

9.3. **Quadratic Lyapunov functions for linear systems.** Consider again

$$y' = Ay. \tag{66}$$

Let Q be a real symmetric positive-definite matrix and define

$$V(y) = y^{\mathsf{T}} Q y. \tag{67}$$

If $\lambda_{\min}(Q)$ and $\lambda_{\max}(Q)$ denote the extreme eigenvalues of Q, then

$$\lambda_{\min}(Q)\|y\|^2 \leq V(y) \leq \lambda_{\max}(Q)\|y\|^2. \tag{68}$$

Consequently, $V(y(t)) \to 0$ if and only if $y(t) \to 0$, and boundedness of $V(y(t))$ is equivalent to boundedness of $y(t)$.

Along a solution of (66),

$$\begin{aligned}\frac{d}{dt}V(y(t)) &= y'(t)^{\mathsf{T}}Qy(t) + y(t)^{\mathsf{T}}Qy'(t)\\ &= y(t)^{\mathsf{T}}\big(A^{\mathsf{T}}Q + QA\big)y(t).\end{aligned} \tag{69}$$

If

$$A^{\mathsf{T}}Q + QA \preceq -\mu I \qquad \text{for some } \mu > 0,$$

then

$$\begin{aligned}\dot{V}(y(t)) &\le -\mu\|y(t)\|^2\\ &\le -\frac{\mu}{\lambda_{\max}(Q)}V(y(t)).\end{aligned}$$

Grönwall's inequality therefore gives

$$V(y(t)) \le e^{-\mu(t-t_0)/\lambda_{\max}(Q)}V(y(t_0)), \tag{70}$$

so the origin is exponentially stable.

9.3.1. *Lyapunov equation.* Choose a symmetric positive-definite matrix C. The matrix equation

$$A^{\mathsf{T}}Q + QA = -C \tag{71}$$

is called the *continuous Lyapunov equation*. Often one simply takes $C = I$ and solves for Q.

Theorem 4 (Lyapunov matrix theorem)**.** *The matrix A is Hurwitz if and only if, for every symmetric $C \succ 0$, equation (71) has a unique symmetric solution $Q \succ 0$.*

Proof. Suppose first that A is Hurwitz. By the stability criterion from the previous lecture, $e^{tA} \to 0$ exponentially. Define

$$Q = \int_0^\infty e^{tA^{\mathsf{T}}}Ce^{tA}\,dt. \tag{72}$$

The integral converges, Q is symmetric, and for $v \neq 0$,

$$v^{\mathsf{T}}Qv = \int_0^\infty \big(e^{tA}v\big)^{\mathsf{T}}C\big(e^{tA}v\big)\,dt > 0.$$

Thus $Q \succ 0$. Moreover,

$$\begin{aligned} A^{\mathsf{T}}Q + QA &= \int_0^\infty \frac{d}{dt}\left(e^{tA^{\mathsf{T}}} C e^{tA}\right) dt \\ &= \left[e^{tA^{\mathsf{T}}} C e^{tA}\right]_{t=0}^{t=\infty} = -C. \end{aligned}$$

If Q_1 and Q_2 are two solutions, their difference D satisfies $A^{\mathsf{T}}D + DA = 0$. Hence $e^{tA^{\mathsf{T}}} D e^{tA} = D$ for all t, while the left-hand side tends to zero. Therefore $D = 0$, proving uniqueness.

Conversely, suppose $Q \succ 0$ and $C \succ 0$ satisfy (71). For a possibly complex eigenpair $Av = \lambda v$, multiplication by the conjugate transpose v^* gives

$$2\,\mathrm{Re}(\lambda) v^* Q v = -v^* C v < 0.$$

Since $v^*Qv > 0$, every eigenvalue has negative real part. Thus A is Hurwitz. □

The integral formula shows the equivalence between the spectral and Lyapunov approaches to linear stability. It also proves the nonlinear stability conclusion locally. Indeed, choose $C = I$ and use the same quadratic function for $y' = Ay + r(y)$. Then

$$\dot{V}(y) = -\|y\|^2 + 2y^{\mathsf{T}} Q r(y) = -\|y\|^2 + o(\|y\|^2),$$

which is negative for all sufficiently small nonzero y.

9.4. **A time-dependent linear system.** Consider

$$y' = A(t)y. \tag{73}$$

For the simple Lyapunov function

$$V(y) = \frac{1}{2} y^{\mathsf{T}} y = \frac{1}{2}\|y\|^2,$$

we have

$$\dot{V}(y(t)) = y(t)^{\mathsf{T}} S(t) y(t), \qquad S(t) = \frac{1}{2}\big(A(t) + A(t)^{\mathsf{T}}\big). \tag{74}$$

Thus $S(t) \preceq 0$ for all $t \geq t_0$ is sufficient for stability, because the Euclidean norm cannot increase. The uniform condition

$$S(t) \preceq -\alpha I, \qquad t \geq t_0, \qquad \alpha > 0, \tag{75}$$

implies

$$\dot{V} \le -2\alpha V, \qquad \|y(t)\| \le e^{-\alpha(t-t_0)}\|y(t_0)\|,$$

and hence uniform exponential stability. Pointwise negative definiteness without a uniform bound does not by itself guarantee decay to zero.

9.5. **Runge–Kutta methods on a linear system.** We now examine the numerical evolution of the constant-coefficient system $y' = Ay$ with a fixed step size τ.

9.5.1. *Forward Euler and an explicit midpoint method.* Forward Euler has one stage,

$$k_1 = Ay_n,$$

and therefore

$$y_{n+1} = y_n + \tau k_1 = (I + \tau A)y_n. \tag{76}$$

The two-stage explicit midpoint method is

$$\begin{aligned} k_1 &= Ay_n, \\ k_2 &= A\left(y_n + \frac{\tau}{2}k_1\right), \\ y_{n+1} &= y_n + \tau k_2. \end{aligned}$$

Consequently,

$$y_{n+1} = \left(I + \tau A + \frac{\tau^2}{2}A^2\right) y_n. \tag{77}$$

Every explicit two-stage Runge–Kutta method of order two has the same update matrix on this linear equation, even though its internal stages may differ.

More generally, recall the Runge–Kutta stability function R from (39). Applied to $y' = Ay$, a Runge–Kutta method gives

$$y_{n+1} = G(\tau A)y_n, \qquad G(\tau A) = R(\tau A). \tag{78}$$

For an explicit s-stage method, R is a polynomial of degree at most s. If that method also has order s, then

$$R(z) = \sum_{j=0}^{s} \frac{z^j}{j!},$$

so $G(\tau A)$ is the degree-s Taylor approximation of $e^{\tau A}$.

9.6. **Stability of a discrete linear evolution.** Consider a general recurrence

$$y_{n+1} = G y_n, \qquad y_n = G^n y_0. \tag{79}$$

The zero solution is stable precisely when $\{G^n\}_{n\geq 0}$ is bounded, and it is asymptotically stable precisely when $G^n \to 0$.

As in the continuous case, write $G = PJP^{-1}$ over $\mathbb{C}$. For a Jordan block

$$J_k = \mu_k I + N_k, \qquad N_k^{q_k} = 0,$$

the binomial theorem gives

$$J_k^n = \sum_{j=0}^{\min(n,q_k-1)} \binom{n}{j} \mu_k^{n-j} N_k^j. \tag{80}$$

The binomial coefficient grows polynomially in n. It is dominated by $|\mu_k|^n$ when $|\mu_k| < 1$, but produces growth on the unit circle when $N_k \neq 0$.

Theorem 5 (Stability criterion for a discrete linear system)**.** *For the recurrence* (79)*:*

(i) *the zero solution is asymptotically stable if and only if*

$$\rho(G) = \max_{\mu\in\sigma(G)} |\mu| < 1;$$

(ii) *the zero solution is stable if and only if* $\rho(G) \leq 1$ *and every eigenvalue with* $|\mu| = 1$ *is semisimple.*

For a Runge–Kutta method, the spectral mapping theorem applied to (78) gives

$$\sigma\big(R(\tau A)\big) = \big\{R(\tau\lambda) : \lambda \in \sigma(A)\big\}.$$

Therefore the numerical solution is asymptotically stable when

$$|R(\tau\lambda)| < 1 \qquad \text{for every } \lambda \in \sigma(A). \tag{81}$$

This motivates the *absolute stability region*

$$\mathcal{S} = \{z \in \mathbb{C} : |R(z)| < 1\}. \tag{82}$$

For example, forward Euler has $R(z) = 1 + z$. If $\lambda < 0$ is real, then $|1 + \tau\lambda| < 1$ is equivalent to

$$0 < \tau < \frac{2}{|\lambda|}.$$

Thus even when the differential equation is asymptotically stable, the step size must place every scaled eigenvalue $\tau\lambda$ inside the method's absolute stability region.

10. September 16, 2026

We study the stability domains of explicit Runge–Kutta methods and use them to choose time steps. The examples show two different obstructions: rapidly decaying modes can force very small steps, and an undamped oscillator can become unstable under discretization. Throughout, A denotes the matrix in the differential equation; $\mathsf{A} = (a_{ij})$ denotes the Runge–Kutta coefficient matrix.

10.1. **Continuous stability and discrete stability.** For a constant matrix A, the solution of $x' = Ax$ is $x(t) = e^{tA}x(0)$. Recall the two spectral criteria:

	asymptotic stability	stability (boundedness)
$x' = Ax$	$\operatorname{Re}\lambda < 0$	$\operatorname{Re}\lambda \le 0$
$x_{n+1} = Gx_n$	$\lvert\mu\rvert < 1$	$\lvert\mu\rvert \le 1$

Here the conditions apply to every $\lambda \in \sigma(A)$ and $\mu \in \sigma(G)$, respectively. In the boundedness column, eigenvalues on the boundary must also be *semisimple*: every associated Jordan block must have size one. Algebraic multiplicity one is sufficient, but is not necessary.

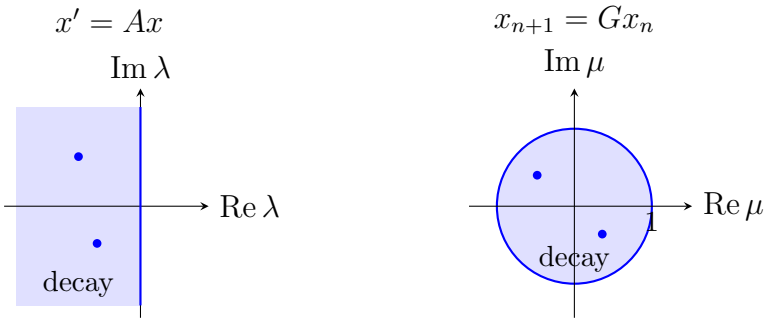


FIGURE 1. The open left half-plane governs continuous decay; the open unit disk governs discrete decay. Boundary eigenvalues require the additional Jordan-block condition for boundedness.

For an explicit RK method applied with a fixed step $\tau > 0$,

$$x_{n+1} = R(\tau A)x_n, \qquad \sigma(R(\tau A)) = \{R(\tau\lambda) : \lambda \in \sigma(A)\}.$$

For example, the eigenvector identity $R(\tau A)v = R(\tau\lambda)v$ follows from $Av = \lambda v$. Triangularizing A proves the full spectral mapping identity even when A is not diagonalizable. Consequently,

$$\rho(R(\tau A)) = \max_{\lambda\in\sigma(A)} |R(\tau\lambda)|,$$

and strict inequality $\rho(R(\tau A)) < 1$ characterizes asymptotic stability of the numerical evolution.

10.2. **The stability domain in the complex plane.**

Definition 8 (Stability domain)**.** *For the scalar test equation $y' = \lambda y$, put $z = \tau\lambda$ and write $y_{n+1} = R(z)y_n$. We use the notation*

$$\mathcal{D} = \{z \in \mathbb{C} : |R(z)| \le 1\}, \qquad \mathcal{S} = \{z \in \mathbb{C} : |R(z)| < 1\}. \tag{83}$$

Thus $\mathcal{D}$ describes bounded scalar iterates, and $\mathcal{S}$ describes decay. For a rational stability function, these sets contain only points where the function is defined.

The handwritten notes use the non-strict convention $\mathcal{D}$; $\mathcal{S}$ agrees with the strict convention used in the previous lecture. In particular, $R(0) = 1$, so $0 \in \mathcal{D}$ but $0 \notin \mathcal{S}$. To reproduce decay of a linear system,

every scaled eigenvalue $\tau\lambda$ must belong to $\mathcal{S}$. Membership in $\mathcal{D}$ alone does not settle a matrix problem with defective boundary eigenvalues.

10.2.1. *Forward Euler.* Here $R_1(z) = 1 + z$. Writing $z = x + iy$,

$$|R_1(z)| \le 1 \quad \Longleftrightarrow \quad (x+1)^2 + y^2 \le 1.$$

The domain is the closed disk of radius one centered at -1. For a negative real eigenvalue,

$$|1 + \tau\lambda| < 1 \quad \Longleftrightarrow \quad 0 < \tau < \frac{2}{|\lambda|}. \tag{84}$$

At the upper endpoint the amplification factor is -1: the scalar solution alternates without decaying.

10.2.2. *Two, three, and four stages.* An explicit two-stage method of order two has

$$R_2(z) = 1 + z + \frac{z^2}{2}.$$

Its boundary can be plotted using

$$\left(1 + x + \frac{x^2 - y^2}{2}\right)^2 + y^2(1+x)^2 = 1.$$

On the negative real axis, $R_2(-r) = 1 - r + r^2/2$ is positive and is less than one exactly when $0 < r < 2$. Thus its real stability interval is the same as forward Euler's, despite its higher order.

The standard three-stage, third-order method and classical RK4 have

$$R_3(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6}, \qquad R_4(z) = 1 + z + \frac{z^2}{2} + \frac{z^3}{6} + \frac{z^4}{24}.$$

Their strict negative-real stability intervals are approximately

$$(-2.512745, 0) \quad \text{and} \quad (-2.785294, 0),$$

respectively. The following plots are computed from $|R_j(z)| = 1$, rather than schematic ovals.

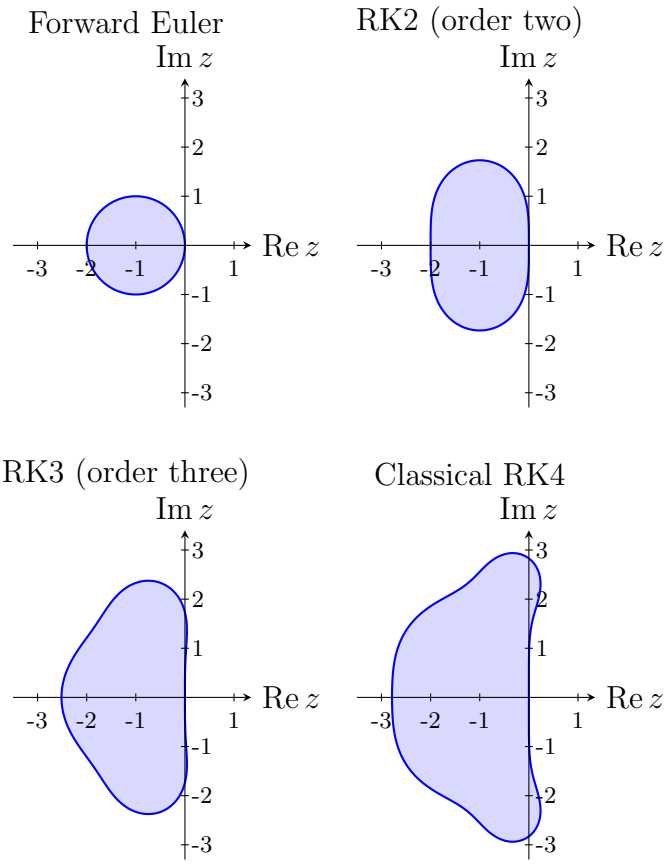


FIGURE 2. Computed stability domains for four standard explicit RK methods, on equal scales. Shading indicates $|R(z)| \le 1$. Higher order changes the shape, but every one of these domains is bounded.

Remark 2 (Stages are not the same as order). An explicit s-stage method has a stability polynomial of degree *at most* s. If its order is p, then

$$R(z) = \sum_{j=0}^{p} \frac{z^j}{j!} + \sum_{j=p+1}^{s} \alpha_j z^j.$$

The degree can be smaller than s, and the coefficients beyond order p depend on the method. The formula $R(z) = \sum_{j=0}^{s} z^j/j!$ applies when the method has order s; it does not describe every s-stage RK method.

10.3. **Why every explicit RK stability domain is bounded.** The statement "the stability domain is finite" means that it is *bounded*, not

that it contains finitely many points. The following completes the polynomial argument indicated in class.

Theorem 6. *Fix any finite number of stages $s \geq 1$. Every consistent explicit s-stage Runge–Kutta method has a bounded stability domain $\mathcal{D}$. In particular, no such method is A-stable.*

Proof. Let $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$. On $y' = \lambda y$, the stage vector satisfies

$$(I - z\mathsf{A})k = \lambda y_n \mathbf{1}, \qquad R(z) = 1 + zb^{\mathsf{T}}(I - z\mathsf{A})^{-1}\mathbf{1}.$$

Since A is strictly lower triangular, $\mathsf{A}^s = 0$. The finite geometric-series identity gives

$$(I - z\mathsf{A})^{-1} = I + z\mathsf{A} + \cdots + z^{s-1}\mathsf{A}^{s-1},$$

for every complex z. Therefore

$$R(z) = 1 + \sum_{j=1}^{s} \big(b^{\mathsf{T}}\mathsf{A}^{j-1}\mathbf{1}\big) z^j. \tag{85}$$

Consistency gives $b^{\mathsf{T}}\mathbf{1} = 1$, so $R'(0) = 1$. In particular, R is not constant. Let its actual degree be m, where $1 \leq m \leq s$, and write

$$R(z) = \alpha_m z^m + \cdots + \alpha_1 z + \alpha_0, \qquad \alpha_m \neq 0.$$

Suppose, for a contradiction, that $\mathcal{D}$ is unbounded. There are $z_n \in \mathcal{D}$ with $|z_n| \to \infty$. On one hand,

$$\left| \frac{R(z_n)}{z_n^m} \right| \leq \frac{1}{|z_n|^m} \longrightarrow 0.$$

On the other hand, division of the polynomial by z_n^m gives

$$\frac{R(z_n)}{z_n^m} = \alpha_m + \sum_{j=0}^{m-1} \frac{\alpha_j}{z_n^{m-j}} \longrightarrow \alpha_m \neq 0,$$

a contradiction. Hence $\mathcal{D}$ is bounded.

For an explicit radius bound, set

$$C = \sum_{j=0}^{m-1} |\alpha_j|, \qquad B = \max\left\{1, \frac{C+1}{|\alpha_m|}\right\}.$$

If $r = |z| > B$, the reverse triangle inequality yields

$$|R(z)| \geq |\alpha_m| r^m - C r^{m-1} = r^{m-1}(|\alpha_m| r - C) > 1.$$

Thus $\mathcal{D} \subseteq \{|z| \leq B\}$. It is also closed by continuity of R, hence compact. Finally, an A-stable method must contain the entire open left half-plane in its stability domain. This is impossible for a bounded set. □

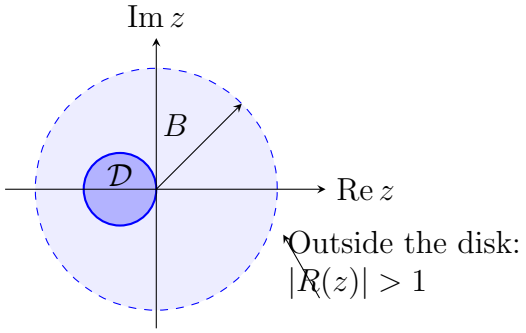


FIGURE 3. The polynomial's leading term dominates in every complex direction. The small region is schematic; the enclosing radius depends on the method's coefficients and need not be sharp.

Remark 3 (Scope of the theorem). The radius may change with s and with the RK coefficients; the theorem does not give one common bound for all explicit methods. Consistency rules out the degenerate constant polynomial $R \equiv 1$, whose non-strict domain would be all of $\mathbb{C}$. Implicit methods generally have rational stability functions and can have unbounded domains, as the next lecture shows.

10.4. **A system with a fast decaying mode.** Consider the example from class,

$$x' = Ax, \qquad A = \begin{bmatrix} 0 & 1 \\ -50 & -51 \end{bmatrix}. \tag{86}$$

Its characteristic polynomial is

$$\det(\lambda I - A) = \lambda^2 + 51\lambda + 50 = (\lambda + 1)(\lambda + 50).$$

With eigenvectors $v_1 = (1, -1)^\mathsf{T}$ and $v_2 = (1, -50)^\mathsf{T}$, the exact solution is

$$x(t) = c_1 e^{-t} v_1 + c_2 e^{-50t} v_2.$$

Both modes decay, but on very different time scales: 1 and 1/50. Forward Euler must satisfy both

$$|1 - \tau| < 1, \qquad |1 - 50\tau| < 1,$$

so its strict stability restriction is

$$\boxed{0 < \tau < \frac{1}{25} = 0.04.} \tag{87}$$

At $\tau = 0.04$, the fast numerical mode has multiplier -1 and fails to decay. Since this matrix is diagonalizable, the endpoint still gives bounded linear iterates. For $\tau > 0.04$, the fast mode is unstable. The second-order, two-stage explicit method has exactly the same restriction because its real stability interval is also $(-2, 0)$.

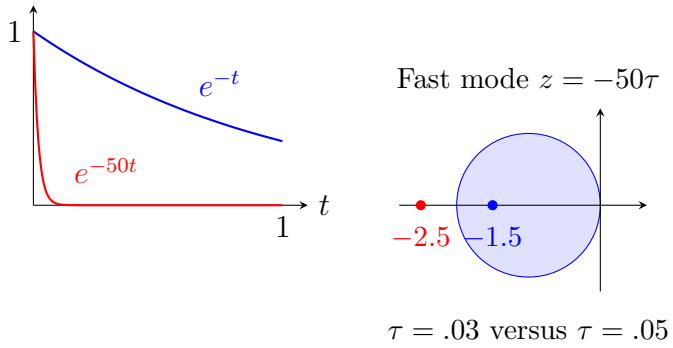


FIGURE 4. The fast exact mode quickly disappears, but it still limits Euler's step size. For $\tau = .03$, its scaled eigenvalue is inside the disk; for $\tau = .05$, it is outside.

This is the mechanism of *stiffness*: stability requires a much smaller step than is needed to resolve the slowly varying solution of interest. Even after the fast exact mode becomes negligible, roundoff and other perturbations can excite its unstable numerical counterpart. Stiffness depends on time scales and the integration interval, not merely on the numerical size of a dimensional eigenvalue.

10.5. **The same restriction near a nonlinear equilibrium.** The nonlinear example in the notes is

$$x' = f(x), \qquad f(x_1, x_2) = \begin{bmatrix} x_2 \\ -51x_2 - 25(x_1^3 - x_1) \end{bmatrix}. \tag{88}$$

At $x_* = (1, 0)$, $f(x_*) = 0$, and

$$Df(x) = \begin{bmatrix} 0 & 1 \\ -25(3x_1^2 - 1) & -51 \end{bmatrix}, \qquad Df(1, 0) = \begin{bmatrix} 0 & 1 \\ -50 & -51 \end{bmatrix}.$$

The equilibrium is locally asymptotically stable because the Jacobian has eigenvalues $-1, -50$.

Euler's map is $\Psi^\tau(x) = x + \tau f(x)$. It fixes x_*, and

$$D\Psi^\tau(x_*) = I + \tau Df(x_*).$$

For $0 < \tau < 1/25$, both multipliers are strictly inside the unit circle, so the discrete equilibrium is locally asymptotically stable. The same conclusion holds for explicit RK2, since the derivative of its map at an equilibrium is $R_2(\tau Df(x_*))$. At the boundary, linearization alone does not determine nonlinear stability.

As an additional illustration, write $q = x_1$, $v = x_2$. The system is a damped particle in the potential

$$U(q) = \frac{25}{4}(q^2 - 1)^2, \qquad E(q, v) = \frac{1}{2}v^2 + U(q), \qquad \frac{dE}{dt} = -51v^2 \leq 0.$$

The two potential wells correspond to $(\pm 1, 0)$; the equilibrium $(0, 0)$ is a saddle because its Jacobian has negative determinant.

10.6. **An oscillator: forward Euler is never stable.** For

$$x' = Ax, \qquad A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \tag{89}$$

the eigenvalues are $\pm i$. Since $A^\mathsf{T} = -A$,

$$\frac{d}{dt}\|x(t)\|_2^2 = 0.$$

The exact trajectories are circles, and the origin is stable but not asymptotically stable. Forward Euler has eigenvalues $1 \pm i\tau$, whose

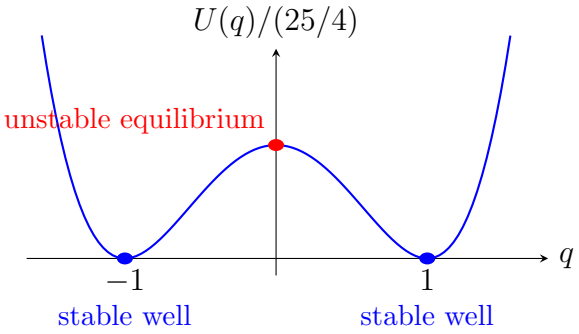


FIGURE 5. The potential of the nonlinear example. Damping decreases mechanical energy; the Jacobian gives the local decay rates near a well.

modulus is $\sqrt{1+\tau^2} > 1$ for every $\tau > 0$. Equivalently,

$$(I+\tau A)^{\mathsf{T}}(I+\tau A) = (1+\tau^2)I, \qquad \|x_n\|_2 = (1+\tau^2)^{n/2}\|x_0\|_2.$$

Thus every positive fixed step causes artificial growth over long integration times. This does not contradict convergence as $\tau \to 0$ on a fixed finite time interval.

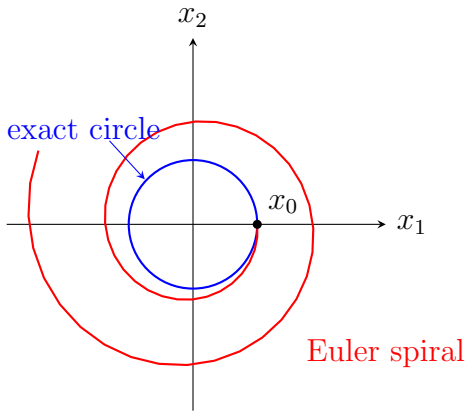


FIGURE 6. Forward Euler on the oscillator, starting at $(1,0)$, with $\tau = .2$ for 50 steps. The exact orbit has radius one; the numerical radius grows by $\sqrt{1+\tau^2}$ at each step.

For comparison, direct calculation gives

$$|R_2(iy)|^2 = 1 + \frac{y^4}{4},$$
$$|R_3(iy)|^2 = 1 + \frac{y^4(y^2-3)}{36},$$
$$|R_4(iy)|^2 = 1 + \frac{y^6(y^2-8)}{576}.$$

Hence RK2 is also unstable on this oscillator for every $\tau > 0$. RK3 and RK4 have bounded iterates for $0 < \tau \le \sqrt{3}$ and $0 < \tau \le 2\sqrt{2}$, respectively. Inside these intervals their amplification factors have modulus less than one, so they introduce numerical damping. Stability alone does not imply preservation of the exact oscillation amplitude or phase.

11. September 18, 2026

We continue with stability domains, then introduce implicit methods whose domains contain the entire left half-plane. We retain the notation x_n, $\tau_n = t_{n+1} - t_n$, and the stage slopes k_i from September 2. For stability calculations the step is fixed, $\tau_n = \tau > 0$. As on September 16, A is the ODE matrix and $\mathsf{A} = (a_{ij})$ is the RK coefficient matrix (denoted by A in the earlier lectures devoted solely to RK coefficients).

11.1. **Step-size restrictions for complex eigenvalues.** For the linear system $x' = Ax$, a Runge–Kutta method gives

$$x_{n+1} = Gx_n, \qquad G = R(\tau A), \qquad \rho(G) = \max_{\lambda \in \sigma(A)} |R(\tau\lambda)|.$$

Strict inequality $\rho(G) < 1$ implies $G^n \to 0$. If $\rho(G) = 1$, boundedness additionally requires every unit-modulus eigenvalue of G to be semisimple.

For forward Euler and $\lambda = a + ib$ with $a < 0$,

$$\begin{aligned} |1 + \tau\lambda|^2 &= (1 + \tau a)^2 + \tau^2 b^2 \\ &= 1 + 2\tau a + \tau^2 |\lambda|^2. \end{aligned}$$

Thus

$$|1+\tau\lambda|<1 \quad\Longleftrightarrow\quad 0<\tau<\frac{-2\,\mathrm{Re}\,\lambda}{|\lambda|^2}. \tag{90}$$

Increasing τ moves $z=\tau\lambda$ outward along a ray; the admissible step ends where that ray leaves the stability disk.

Example 16 (A damped oscillation)**.** *The matrix in class is*

$$A=\begin{bmatrix}1 & 2\\ -4 & -3\end{bmatrix}, \qquad \det(\lambda I-A)=\lambda^2+2\lambda+5.$$

Its eigenvalues are $\lambda_{1,2}=-1\pm 2i$*. Euler's condition is*

$$|1+\tau(-1\pm 2i)|^2=(1-\tau)^2+4\tau^2=1-2\tau+5\tau^2<1.$$

Therefore the numerical solution decays exactly when

$$\boxed{0<\tau<\frac{2}{5}.}$$

At $\tau=2/5$*, the multipliers are* $3/5\pm(4/5)i$*. They are distinct and have modulus one, so the iterates are bounded but do not decay. For larger steps the origin is unstable.*

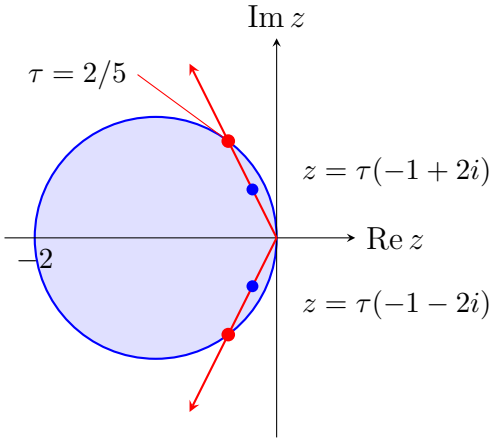


FIGURE 7. The scaled eigenvalues travel along conjugate rays. The points for $\tau=1/5$ lie inside Euler's disk; the red points for $\tau=2/5$ lie on its boundary.

For stiff equations, these restrictions can require many steps even when the solution of interest changes slowly. This motivates methods with no step-size restriction for decay of the linear test equation.

11.2. **Backward Euler.** Backward Euler evaluates the slope at the new time and state:

$$x_{n+1} = x_n + \tau f(t_{n+1}, x_{n+1}). \tag{91}$$

For $f(t, x) = Ax$,

$$(I - \tau A)x_{n+1} = x_n, \qquad G_{\mathrm{BE}} = (I - \tau A)^{-1}.$$

Why is the inverse defined for every $\tau > 0$ if A is Hurwitz? If $I - \tau A$ were singular, there would be a nonzero v with $Av = \tau^{-1}v$. This would give a positive real eigenvalue of A, a contradiction.

The rational spectral mapping identity gives

$$\sigma(G_{\mathrm{BE}}) = \left\{ \frac{1}{1 - \tau\lambda} : \lambda \in \sigma(A) \right\}. \tag{92}$$

For completeness, if $B = I - \tau A$ is invertible, its eigenvalues are $1 - \tau\lambda$, and those of B^{-1} are their reciprocals. This also follows by triangularizing A; no diagonalizability assumption is needed.

Thus the stability function and its non-strict domain are

$$R_{\mathrm{BE}}(z) = \frac{1}{1 - z}, \qquad \mathcal{D}_{\mathrm{BE}} = \{z : |z - 1| \geq 1\}. \tag{93}$$

The domain is the exterior, including the boundary, of the unit disk centered at 1. The pole $z = 1$ is excluded automatically. For $\operatorname{Re} z < 0$,

$$|1 - z|^2 = 1 - 2\operatorname{Re} z + |z|^2 > 1,$$

so $|R_{\mathrm{BE}}(z)| < 1$. Therefore, for every Hurwitz matrix A and every fixed $\tau > 0$, backward Euler is asymptotically stable. This is a statement about stability; resolving a solution accurately can still require a small step.

11.3. **The implicit trapezoidal method.** The trapezoidal rule averages the old and new slopes:

$$x_{n+1} = x_n + \frac{\tau}{2}\big(f(t_n, x_n) + f(t_{n+1}, x_{n+1})\big). \tag{94}$$

This is the *implicit* trapezoidal method. Heun's method, the explicit trapezoidal method from September 2, instead evaluates the second slope at a forward Euler prediction.

For $x' = Ax$, rearranging gives

$$\left(I - \frac{\tau}{2}A\right) x_{n+1} = \left(I + \frac{\tau}{2}A\right) x_n, \qquad G_{\mathrm{TR}} = \left(I - \frac{\tau}{2}A\right)^{-1}\left(I + \frac{\tau}{2}A\right).$$

If A is Hurwitz, the inverse exists for every $\tau > 0$ by the same argument as for backward Euler. Its eigenvalues are

$$\sigma(G_{\mathrm{TR}}) = \left\{ \frac{1 + \tau\lambda/2}{1 - \tau\lambda/2} : \lambda \in \sigma(A) \right\}.$$

Hence

$$R_{\mathrm{TR}}(z) = \frac{1 + z/2}{1 - z/2} = \frac{2 + z}{2 - z}. \tag{95}$$

Writing $z = x + iy$,

$$\begin{aligned} |R_{\mathrm{TR}}(z)| \le 1 \quad &\Longleftrightarrow \quad |2 + z|^2 \le |2 - z|^2 \\ &\Longleftrightarrow \quad (2 + x)^2 + y^2 \le (2 - x)^2 + y^2 \\ &\Longleftrightarrow \quad x \le 0. \end{aligned}$$

Thus

$$\mathcal{D}_{\mathrm{TR}} = \{z : \operatorname{Re} z \le 0\}, \qquad \mathcal{S}_{\mathrm{TR}} = \{z : \operatorname{Re} z < 0\}. \tag{96}$$

The pole $z = 2$ lies outside these sets. The trapezoidal method is also asymptotically stable for every positive fixed step when A is Hurwitz.

Definition 9 (A-stability)**.** *A method is* A-stable *if its stability function is defined and satisfies* $|R(z)| \le 1$ *throughout the open left half-plane.*

Backward Euler and the trapezoidal method are A-stable. Theorem 6 shows that no consistent explicit RK method with finitely many stages can be A-stable.

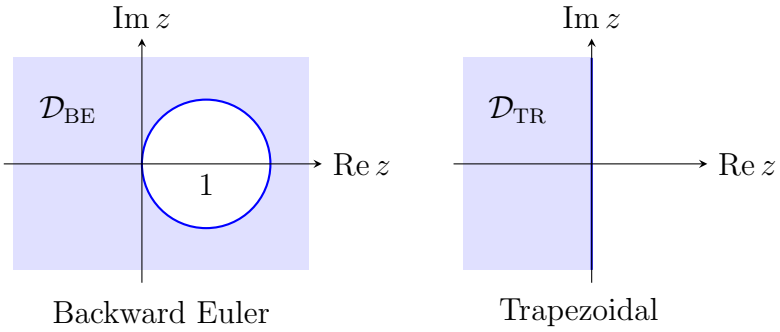


FIGURE 8. The shaded stability domains extend to infinity. Both contain the entire left half-plane; backward Euler also damps some modes in the right half-plane, which is not faithful to the growth of those exact modes.

11.4. **Oscillations: growth, damping, or constant amplitude.** Return to the skew-symmetric oscillator matrix $A = \left[\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right]$ from (89). The exact norm is constant. The scalar amplification factors on the imaginary axis satisfy

method	$\lvert R(i\tau)\rvert$	amplitude behavior
forward Euler	$\sqrt{1+\tau^2}$	growth
backward Euler	$(1+\tau^2)^{-1/2}$	decay
trapezoidal	1	constant

For backward Euler,

$$\|x_n\|_2 = (1+\tau^2)^{-n/2}\|x_0\|_2;$$

the damping is artificial for this conservative problem.

For the trapezoidal method, put $q = \tau/2$. Since $A^2 = -I$,

$$G_{\mathrm{TR}} = \frac{(1-q^2)I + 2qA}{1+q^2}, \qquad G_{\mathrm{TR}}^{\mathsf{T}} G_{\mathrm{TR}} = I.$$

Thus the numerical update is an orthogonal rotation and preserves $\|x_n\|_2$ exactly. More generally, the same orthogonality follows for every real skew-symmetric matrix A, since the factors $I \pm qA$ commute.

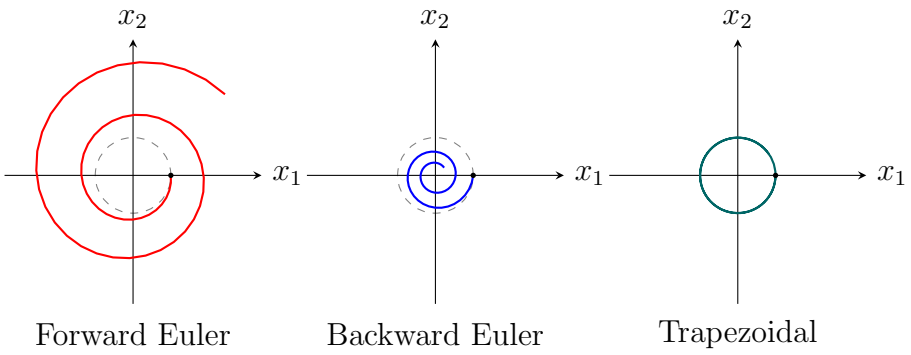


FIGURE 9. The same oscillator and initial value $(1, 0)$, with $\tau = .2$ for 60 steps in each panel, on identical scales. The dashed circle is the exact orbit. Trapezoidal vertices remain on the circle; the straight segments only join successive iterates.

Amplitude preservation does not imply exact phase. On the scalar mode $y' = iy$,

$$R_{\mathrm{TR}}(i\tau) = \exp\left(2i \arctan \frac{\tau}{2}\right), \qquad 2\arctan\frac{\tau}{2} = \tau - \frac{\tau^3}{12} + O(\tau^5).$$

The exact phase advance is τ. The numerical frequency is

$$\frac{2}{\tau}\arctan\frac{\tau}{2}.$$

For the real oscillator drawn here, the rotation is clockwise, with this same angle magnitude. Purely imaginary eigenvalues alone do not imply conservation of the Euclidean norm for an arbitrary matrix; skew-symmetry is the property used in this example.

11.5. **Damping very stiff modes: an additional comparison.** Both implicit methods allow arbitrary steps for linear stability, but they behave differently when $z = -r$ with $r \gg 1$:

$$R_{\mathrm{BE}}(-r) = \frac{1}{1+r} \longrightarrow 0, \qquad R_{\mathrm{TR}}(-r) = \frac{1 - r/2}{1 + r/2} \longrightarrow -1.$$

The exact factor e^{-r} is very small. Backward Euler strongly damps such a mode, while the trapezoidal method can produce slowly decaying sign alternations when the step is large compared with the mode's decay time.

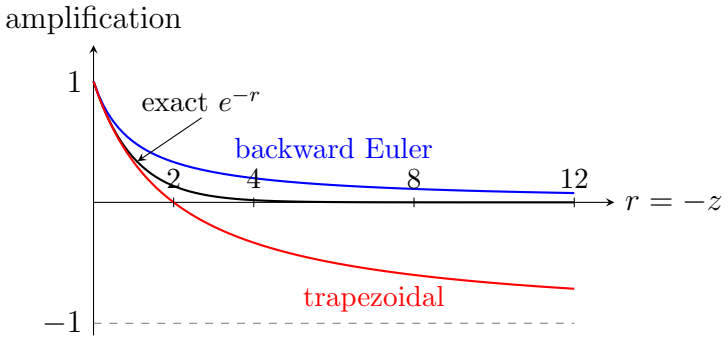


FIGURE 10. Signed amplification factors along the negative real axis. The trapezoidal factor changes sign at $r = 2$ and approaches -1; backward Euler's factor approaches zero.

An A-stable method is called *L-stable* if in addition $R(z) \to 0$ as $|z| \to \infty$ in the left half-plane. Thus backward Euler is L-stable and the trapezoidal method is not. This additional notion explains their different behavior on very fast decaying modes.

11.6. **Implicit Runge–Kutta methods and their tableaux.** The stage and update formulas are the same as in (26)–(27):

$$k_i = f\left(t_n + c_i\tau, x_n + \tau \sum_{j=1}^{s} a_{ij}k_j\right), \qquad i = 1, \dots, s,$$

$$x_{n+1} = x_n + \tau \sum_{i=1}^{s} b_i k_i.$$

For an implicit method, the stage equations need not be solvable by successive explicit evaluations: a stage may depend on itself or on other unknown stages. They generally require solving algebraic systems, nonlinear if f is nonlinear. The internal consistency relation remains $c = \mathsf{A1}$.

11.6.1. *Backward Euler as a one-stage RK method.* Take $a_{11} = b_1 = c_1 = 1$:

$$\begin{array}{c|c} 1 & 1 \\ \hline & 1 \end{array}.$$

Then

$$k_1 = f(t_n + \tau, x_n + \tau k_1), \qquad x_{n+1} = x_n + \tau k_1,$$

which is exactly (91). The coefficient $a_{11} = 1$ makes the sole stage implicit.

11.6.2. *Trapezoidal rule as a two-stage RK method.* Its tableau is

$$\begin{array}{c|cc} 0 & 0 & 0 \\ 1 & \frac{1}{2} & \frac{1}{2} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}, \qquad \mathsf{A} = \begin{bmatrix} 0 & 0 \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix}.$$

Indeed,

$$\begin{aligned} k_1 &= f(t_n, x_n), \\ k_2 &= f\left(t_n + \tau, x_n + \frac{\tau}{2}k_1 + \frac{\tau}{2}k_2\right), \\ x_{n+1} &= x_n + \frac{\tau}{2}(k_1 + k_2). \end{aligned}$$

The argument of f in the second stage is x_{n+1}, so this recovers (94). Its weights satisfy $b^{\mathsf{T}}\mathbf{1} = 1$ and $b^{\mathsf{T}}c = 1/2$, giving order two by the earlier order conditions. In contrast, backward Euler has $b^{\mathsf{T}}c = 1$, so it has only order one.

11.7. **Derivation of the general RK stability function.** On $y' = \lambda y$, each stage satisfies

$$k_i = \lambda y_n + z \sum_{j=1}^{s} a_{ij} k_j, \qquad z = \tau\lambda.$$

Collecting the slopes into $k = (k_1, \dots, k_s)^{\mathsf{T}}$ gives

$$(I - z\mathsf{A})k = \lambda y_n \mathbf{1}.$$

Whenever $I - z\mathsf{A}$ is invertible, substitution into the update produces

$$\begin{aligned} y_{n+1} &= y_n + \tau b^{\mathsf{T}} k \\ &= \left(1 + z b^{\mathsf{T}} (I - z\mathsf{A})^{-1} \mathbf{1}\right) y_n. \end{aligned}$$

Thus the formula from September 4 applies to both explicit and implicit RK schemes:

$$\boxed{R(z) = 1 + zb^{\mathsf{T}}(I - z\mathsf{A})^{-1}\mathbf{1}.} \tag{97}$$

For implicit methods the inverse is generally a rational matrix function, so R need not be a polynomial. The determinant lemma also gives

$$R(z) = \frac{\det(I - z\mathsf{A} + z\mathbf{1}b^{\mathsf{T}})}{\det(I - z\mathsf{A})}.$$

The numerator and denominator have degree at most s, before possible cancellations. At a singular stage matrix, the original stage equations must still be checked; a removable singularity in a simplified scalar formula does not by itself make the stages uniquely solvable.

For backward Euler this yields $R(z) = 1 + z/(1 - z) = 1/(1 - z)$. For the trapezoidal tableau, solve

$$(I - z\mathsf{A})w = \mathbf{1} \quad \Longrightarrow \quad w_1 = 1, \qquad w_2 = \frac{1 + z/2}{1 - z/2}.$$

Then $R(z) = 1 + z(w_1 + w_2)/2 = (1 + z/2)/(1 - z/2)$, as above. Implicitness permits A-stability but does not guarantee it; the actual rational stability function must be examined.

11.8. **Connection with collocation.** The lecture lists collocation as the next topic. The following short construction explains its connection with the implicit RK formulas. Choose distinct nodes $c_1, \dots, c_s \in [0, 1]$ and let u be a polynomial of degree at most s on one time step such that

$$u(t_n) = x_n, \qquad u'(t_n + c_i\tau) = f(t_n + c_i\tau, u(t_n + c_i\tau)).$$

Let $\ell_j(\theta)$ be the Lagrange basis at these nodes and set $k_j = u'(t_n + c_j\tau)$. Interpolation of u', followed by integration, gives

$$u(t_n + \theta\tau) = x_n + \tau \sum_{j=1}^{s} \left(\int_0^{\theta} \ell_j(\xi)\, d\xi \right) k_j.$$

Evaluating at the nodes and at $\theta = 1$ produces the RK coefficients

$$a_{ij} = \int_0^{c_i} \ell_j(\xi)\, d\xi, \qquad b_j = \int_0^1 \ell_j(\xi)\, d\xi. \tag{98}$$

The single node $c_1 = 1$ gives backward Euler. The two nodes $0, 1$ give $\ell_1(\theta) = 1 - \theta$ and $\ell_2(\theta) = \theta$, hence exactly the trapezoidal tableau above.

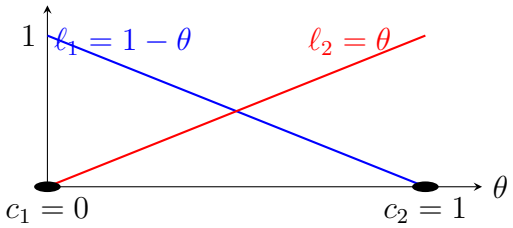


FIGURE 11. The endpoint Lagrange basis for trapezoidal collocation. Each basis function has integral $1/2$, giving the two equal weights; its integral up to each node gives the corresponding tableau row.

12. SEPTEMBER 21, 2026

We continue the study of implicit Runge–Kutta methods, derive their stability functions, and develop the collocation construction outlined on September 18. The main connection is that integrating an interpolating polynomial produces both the RK coefficients and an associated quadrature rule.

We retain x_n, $\tau_n = t_{n+1} - t_n$, and the stage slopes k_i. On a single step, write $\tau = \tau_n$. As in the previous lecture, $\mathsf{A} = (a_{ij})$ denotes the RK coefficient matrix.

12.1. **Implicit Runge–Kutta methods and stability functions.**

An s-stage RK method has the form

$$k_i = f\left(t_n + c_i\tau, x_n + \tau \sum_{j=1}^{s} a_{ij} k_j\right), \qquad i = 1, \dots, s, \tag{99}$$

$$x_{n+1} = x_n + \tau \sum_{i=1}^{s} b_i k_i. \tag{100}$$

The coefficients are arranged in the Butcher tableau

$$\begin{array}{c|c} c & \mathsf{A} \\ \hline & b^{\mathsf{T}} \end{array}, \qquad c = \begin{bmatrix} c_1 \\ \vdots \\ c_s \end{bmatrix}, \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_s \end{bmatrix}.$$

For an implicit method, the stage equations generally form a coupled algebraic system. They are nonlinear when f is nonlinear.

Apply the method to the scalar test equation

$$x' = \lambda x, \qquad \lambda \in \mathbb{C}, \qquad z = \tau\lambda.$$

The stage equations become

$$k_i = \lambda x_n + z \sum_{j=1}^{s} a_{ij} k_j.$$

With $k = (k_1, \dots, k_s)^{\mathsf{T}}$ and $\mathbf{1} = (1, \dots, 1)^{\mathsf{T}}$, this is

$$(I - z\mathsf{A})k = \lambda x_n \mathbf{1}.$$

Whenever $I - z\mathsf{A}$ is invertible,

$$k = \lambda x_n (I - z\mathsf{A})^{-1}\mathbf{1}.$$

Substitution into the update gives

$$x_{n+1} = R(z)x_n, \qquad \boxed{R(z) = 1 + zb^{\mathsf{T}}(I - z\mathsf{A})^{-1}\mathbf{1}.} \tag{101}$$

Thus an implicit RK method generally has a rational stability function. For an explicit method, $\mathsf{A}^s = 0$, so

$$(I - z\mathsf{A})^{-1} = I + z\mathsf{A} + \cdots + z^{s-1}\mathsf{A}^{s-1},$$

and R is a polynomial of degree at most s.

Keeping the distinction from the previous lectures, the strict absolute stability region and the non-strict stability domain are

$$\mathcal{S} = \{z \in \mathbb{C} : I - z\mathsf{A} \text{ is invertible, } |R(z)| < 1\},$$
$$\mathcal{D} = \{z \in \mathbb{C} : I - z\mathsf{A} \text{ is invertible, } |R(z)| \le 1\}.$$

For a fixed step, $x_n = R(z)^n x_0$. Hence points in $\mathcal{S}$ give decay to zero, while points with $|R(z)| = 1$ give bounded scalar iterates without

decay. The invertibility condition ensures that the stages themselves are uniquely defined.

12.2. **The collocation polynomial.** Consider a system $x' = f(t, x)$, where the solution takes values in $\mathbb{R}^d$. Choose distinct nodes $c_1, \dots, c_s \in [0, 1]$. Let $\mathcal{P}_m$ denote the scalar polynomials of degree at most m.

Definition 10 (Collocation on one time step)**.** *Seek a vector polynomial $u \in (\mathcal{P}_s)^d$ on $[t_n, t_{n+1}]$ satisfying*

$$u(t_n) = x_n, \tag{102}$$

$$u'(t_n + c_i\tau) = f\big(t_n + c_i\tau, u(t_n + c_i\tau)\big), \qquad i = 1, \dots, s. \tag{103}$$

The next numerical value is

$$x_{n+1} = u(t_{n+1}).$$

The polynomial agrees with the initial value and satisfies the differential equation at the chosen collocation points. It need not satisfy the differential equation between those points. The initial condition and the s collocation conditions give $(s + 1)d$ scalar equations for the $(s+1)d$ polynomial coefficients. For smooth f, the corresponding stage equations have a unique local solution near $\tau = 0$; arbitrary large steps need not have this property.

12.3. **Lagrange interpolation and the RK coefficients.** The Lagrange basis for the nodes is

$$\ell_j(\theta) = \prod_{\substack{m=1 \\ m \neq j}}^{s} \frac{\theta - c_m}{c_j - c_m}, \qquad \ell_j(c_i) = \delta_{ij}. \tag{104}$$

Each ℓ_j has degree at most $s - 1$. Given scalar data $\varphi(c_j)$, the polynomial

$$P(\theta) = \sum_{j=1}^{s} \varphi(c_j)\ell_j(\theta)$$

interpolates φ at all the nodes. In particular, if $\varphi \in \mathcal{P}_{s-1}$, then $P = \varphi$ identically.

Define the collocation slopes by

$$k_j = u'(t_n + c_j\tau) = f\big(t_n + c_j\tau, u(t_n + c_j\tau)\big).$$

Since u' has degree at most $s-1$, its interpolation is exact:

$$u'(t_n + \theta\tau) = \sum_{j=1}^{s} k_j \ell_j(\theta). \tag{105}$$

Integrating and using the initial condition gives

$$\begin{aligned} u(t_n + \theta\tau) - x_n &= \tau \int_0^\theta u'(t_n + \xi\tau)\, d\xi \\ &= \tau \sum_{j=1}^{s} \left(\int_0^\theta \ell_j(\xi)\, d\xi \right) k_j. \end{aligned} \tag{106}$$

The factor τ comes from the change of variables $t = t_n + \xi\tau$.

At $\theta = c_i$, equation (106) yields

$$u(t_n + c_i\tau) = x_n + \tau \sum_{j=1}^{s} a_{ij} k_j, \qquad a_{ij} = \int_0^{c_i} \ell_j(\theta)\, d\theta.$$

Inserting these stage values into (103) gives exactly the RK stage equations (99). At $\theta = 1$, we obtain

$$x_{n+1} = x_n + \tau \sum_{j=1}^{s} b_j k_j, \qquad b_j = \int_0^1 \ell_j(\theta)\, d\theta.$$

Consequently, the collocation coefficients are

$$\boxed{a_{ij} = \int_0^{c_i} \ell_j(\theta)\, d\theta, \qquad b_j = \int_0^1 \ell_j(\theta)\, d\theta.} \tag{107}$$

Conversely, slopes satisfying this RK system define a collocation polynomial through (106). Thus the two formulations are equivalent.

Because $\sum_j \ell_j(\theta) = 1$, the coefficients automatically satisfy

$$\sum_{j=1}^{s} a_{ij} = c_i, \qquad \sum_{j=1}^{s} b_j = 1.$$

These are the internal consistency relation $c = \mathsf{A}\mathbf{1}$ and the first-order condition $b^{\mathsf{T}}\mathbf{1} = 1$.

Collocation generally produces implicit RK methods: the integrated basis functions couple the unknown stages. The one-node choice $c_1 = 0$ is an exception, giving forward Euler. The choices $c_1 = 1$ and $(c_1, c_2) = (0, 1)$ recover backward Euler and the implicit trapezoidal method, respectively, as on September 18.

12.4. **The associated quadrature rule.** Integrating the Lagrange interpolant of a function φ gives the interpolatory quadrature rule

$$\int_0^1 \varphi(\theta)\, d\theta \approx \sum_{j=1}^{s} b_j \varphi(c_j), \qquad b_j = \int_0^1 \ell_j(\theta)\, d\theta. \tag{108}$$

The c_j are the quadrature nodes and the b_j are its weights. Write the quadrature error as

$$E[\varphi] = \int_0^1 \varphi(\theta)\, d\theta - \sum_{j=1}^{s} b_j \varphi(c_j).$$

We use E here to distinguish the error functional from the stability function $R(z)$.

Definition 11 (Degree of precision)**.** *A quadrature rule has* degree of precision p *if*

$$E[\varphi] = 0 \quad \textit{for every } \varphi \in \mathcal{P}_p,$$

but $E[\varphi] \neq 0$ *for at least one* $\varphi \in \mathcal{P}_{p+1}$.

Every interpolatory rule with these s distinct nodes has $p \geq s - 1$. Indeed, for $\varphi \in \mathcal{P}_{s-1}$,

$$\varphi(\theta) = \sum_{j=1}^{s} \varphi(c_j) \ell_j(\theta),$$

and integrating proves exactness. Equivalently, exactness through degree p is expressed by the moment conditions

$$\sum_{j=1}^{s} b_j c_j^m = \frac{1}{m+1}, \qquad m = 0, \dots, p. \tag{109}$$

For $m = 0$, the left-hand side means $\sum_j b_j$.

The largest possible degree of precision for s real nodes is $2s-1$. To see the upper bound, consider

$$q(\theta) = \prod_{j=1}^{s} (\theta - c_j)^2.$$

This polynomial has degree $2s$, vanishes at every node, and has strictly positive integral on $[0,1]$. Therefore no such rule can be exact for every polynomial of degree $2s$. Gaussian quadrature attains the bound $p = 2s-1$ with a suitable choice of nodes. Its use in RK methods is the topic announced for the next lecture.

12.5. **Quadrature precision and collocation order.**

Theorem 7 (Order of a collocation method)**.** *Suppose the interpolatory quadrature rule associated with the collocation nodes has degree of precision p. Then the corresponding collocation RK method has order $p+1$, assuming sufficient smoothness of f and the solution.*

Here p denotes quadrature precision; the RK order is $p+1$. Thus a step started from the exact solution satisfies

$$x(t_{n+1}) - u(t_{n+1}) = O(\tau^{p+2}).$$

Under the usual smoothness and Lipschitz assumptions, this leads to global error $O(h^{p+1})$ on a fixed finite interval, where $h = \max_n \tau_n$. The order statement concerns the values at step endpoints; it does not assert the same error order at every point of the collocation polynomial.

The role of quadrature is especially transparent when $f(t,x) = g(t)$ is independent of x. Then

$$x_{n+1} = x_n + \tau \sum_{j=1}^{s} b_j g(t_n + c_j \tau),$$

whereas the exact increment is

$$x(t_{n+1}) - x(t_n) = \tau \int_0^1 g(t_n + \theta\tau)\, d\theta.$$

Taylor expansion of g and the moment conditions (109) give a one-step error of $O(\tau^{p+2})$. The theorem extends this order conclusion to general nonlinear collocation equations.

Example 17 (One collocation node)**.** *With $s = 1$ and $c_1 = c$, the basis polynomial is $\ell_1 = 1$, so the tableau is*

$$\begin{array}{c|c} c & c \\ \hline & 1 \end{array}.$$

The quadrature rule is $\int_0^1 \varphi \approx \varphi(c)$. It is exact for linear polynomials precisely when $c = 1/2$. Thus the midpoint choice gives the second-order implicit midpoint method

$$x_{n+1} = x_n + \tau f\left(t_n + \frac{\tau}{2}, \frac{x_n + x_{n+1}}{2}\right).$$

The endpoint choices $c = 0$ and $c = 1$ have precision zero and give the first-order forward and backward Euler methods.

Example 18 (Endpoint collocation)**.** *For $c_1 = 0$, $c_2 = 1$, the weights are $b_1 = b_2 = 1/2$. The quadrature rule is exact for constants and linear polynomials, but*

$$\frac{1}{2}(0^2 + 1^2) = \frac{1}{2} \neq \frac{1}{3} = \int_0^1 \theta^2 \, d\theta.$$

Hence $p = 1$, and the associated trapezoidal collocation method has order two.

For Gaussian nodes, $p = 2s - 1$, so the resulting s-stage Gauss collocation method has order $2s$. In particular, one Gauss node gives order two and two Gauss nodes give order four. The placement of the nodes, not just their number, determines the quadrature precision and hence the collocation order.

13. September 23, 2026

We explain why quadrature precision determines the endpoint order of a collocation method, then construct Gauss Runge–Kutta methods. Exactness of Gaussian quadrature also gives preservation of quadratic

first integrals and a direct proof of A-stability. We retain $\tau = \tau_n$, the stage slopes k_i, and the RK matrix A; A is reserved for an ODE matrix.

13.1. **Collocation and quadrature error.** Recall that the collocation polynomial $u \in (\mathcal{P}_s)^d$ satisfies

$$\begin{aligned} u(t_n) &= x_n, \\ u'(t_n + c_i\tau) &= f\big(t_n + c_i\tau, u(t_n + c_i\tau)\big), \qquad i = 1, \dots, s, \\ x_{n+1} &= u(t_n + \tau). \end{aligned}$$

The associated RK coefficients are

$$a_{ij} = \int_0^{c_i} \ell_j(\theta)\, d\theta, \qquad b_j = \int_0^1 \ell_j(\theta)\, d\theta,$$

where ℓ_j is the Lagrange basis from (104).

On the physical time interval, define the quadrature error functional

$$E_{n,\tau}[\varphi] = \int_{t_n}^{t_n+\tau} \varphi(t)\, dt - \tau \sum_{j=1}^{s} b_j \varphi(t_n + c_j\tau). \tag{110}$$

This is a linear functional. As on September 21, E denotes quadrature error, while $R(z)$ denotes a stability function.

Lemma 2 (Scaling of quadrature error)**.** *Suppose the quadrature rule has degree of precision p. For $\varphi \in C^{p+1}([t_n, t_n + \tau])$,*

$$|E_{n,\tau}[\varphi]| \le C\tau^{p+2} \max_{t_n \le t \le t_n+\tau} |\varphi^{(p+1)}(t)|,$$

where C depends only on the fixed nodes and weights.

Proof. Expand $\varphi(t_n + \tau\theta)$ through degree p:

$$\varphi(t_n + \tau\theta) = \sum_{m=0}^{p} \frac{\varphi^{(m)}(t_n)}{m!} \tau^m \theta^m + r(\theta),$$

with

$$|r(\theta)| \le \frac{\tau^{p+1}}{(p+1)!} \max |\varphi^{(p+1)}|, \qquad 0 \le \theta \le 1.$$

Exactness cancels every displayed polynomial term. The remaining integral and weighted sum are bounded by $\tau(1 + \sum_j |b_j|) \max |r|$, proving the estimate. □

The same estimate holds componentwise for vector-valued functions. If the integrand itself depends on τ, the derivative bound must be uniform as $\tau \to 0$.

13.2. **The collocation defect and variation of constants.** For the local error analysis, let $x(t)$ be the exact solution started from $x(t_n) = x_n$. Define the defect

$$h(t) = u'(t) - f(t, u(t)). \tag{111}$$

Then the polynomial solves a perturbed differential equation,

$$u' = f(t, u) + h(t), \qquad u(t_n) = x_n,$$

and the collocation conditions say exactly that

$$h(t_n + c_i\tau) = 0, \qquad i = 1, \dots, s. \tag{112}$$

For the linear problem $x' = Ax$, variation of constants gives

$$u(t) = e^{(t-t_n)A}x_n + \int_{t_n}^{t} e^{(t-\eta)A}h(\eta)\, d\eta.$$

Thus, writing $T = t_n + \tau$,

$$x(T) - u(T) = -\int_{t_n}^{T} e^{(T-t)A}h(t)\, dt. \tag{113}$$

The minus sign follows from defining $h = u' - f(t, u)$ and taking the local error to be exact minus numerical, as in (19).

For a nonlinear equation, the analogous identity uses the exact flow $\Phi^{T,t}(y)$, namely the solution at time T started from y at time t. Put

$$M(T, t) = D_y\Phi^{T,t}(u(t)).$$

The flow identity $\partial_t\Phi^{T,t}(y) = -D_y\Phi^{T,t}(y)f(t, y)$ gives

$$\frac{d}{dt}\Phi^{T,t}(u(t)) = M(T, t)h(t).$$

Integrating from t_n to T yields the nonlinear variation-of-constants formula

$$x(T) - u(T) = -\int_{t_n}^{T} M(T, t)h(t)\, dt. \tag{114}$$

Set $\varphi(t) = M(T,t)h(t)$. By (112), every quadrature sample vanishes, so

$$x(T) - u(T) = -E_{n,\tau}[\varphi].$$

For sufficiently smooth f, on the local branch of collocation solutions for small τ, the required derivatives of φ remain bounded. The quadrature error estimate therefore gives

$$\boxed{x(t_n+\tau) - u(t_n+\tau) = O(\tau^{p+2}).} \tag{115}$$

This proves the order $p+1$ conclusion stated on September 21. The extra accuracy at the endpoint comes from cancellation in the integral of the defect; merely bounding $\|h\|$ would miss that cancellation.

13.3. **Gauss collocation.** Gaussian quadrature with s nodes has degree of precision $2s-1$, the largest possible value. Its nodes are the roots of the shifted Legendre polynomial of degree s:

$$L_s(\theta) = P_s(2\theta - 1), \qquad 0 < c_1 < \cdots < c_s < 1,$$

where P_s is the Legendre polynomial on $[-1,1]$. Equivalently, the monic nodal polynomial $\pi_s(\theta) = \prod_{j=1}^{s}(\theta - c_j)$ is orthogonal to $\mathcal{P}_{s-1}$ on $[0,1]$:

$$\int_0^1 \pi_s(\theta) q(\theta)\, d\theta = 0, \qquad q \in \mathcal{P}_{s-1}.$$

To see the resulting precision, divide any $v \in \mathcal{P}_{2s-1}$ as $v = \pi_s q + r$, with $q, r \in \mathcal{P}_{s-1}$. The integral of $\pi_s q$ vanishes by orthogonality, and all its quadrature samples vanish as well. The interpolatory rule integrates r exactly. Hence it integrates v exactly.

The nodes and weights satisfy

$$c_j = 1 - c_{s+1-j}, \qquad b_j = b_{s+1-j} > 0. \tag{116}$$

Positivity follows directly from exactness applied to ℓ_j^2, whose degree is $2s-2$:

$$b_j = \sum_{i=1}^{s} b_i \ell_j(c_i)^2 = \int_0^1 \ell_j(\theta)^2\, d\theta > 0.$$

Collocation at these nodes gives the *Gauss RK method* with s stages. The quadrature precision is $2s-1$, so (115) gives

$$\text{RK order} = 2s, \qquad \text{local error} = O(\tau^{2s+1}).$$

13.3.1. *One stage: implicit midpoint.* For $s=1$, $c_1 = 1/2$, $b_1 = 1$, and $a_{11} = 1/2$:

$$\begin{array}{c|c} \frac{1}{2} & \frac{1}{2} \\ \hline & 1 \end{array}.$$

Thus

$$k_1 = f\left(t_n + \frac{\tau}{2}, x_n + \frac{\tau}{2}k_1\right), \qquad x_{n+1} = x_n + \tau k_1.$$

Equivalently,

$$x_{n+1} = x_n + \tau f\left(t_n + \frac{\tau}{2}, \frac{x_n + x_{n+1}}{2}\right).$$

This method has order two. Its stability function is

$$R_{\mathrm{G},1}(z) = \frac{1 + z/2}{1 - z/2}.$$

It agrees with the trapezoidal stability function on the linear test equation, although the two methods differ for general nonlinear f.

13.3.2. *Two-stage Gauss method.* The roots of $L_2(\theta) = 6\theta^2 - 6\theta + 1$ are

$$c_1 = \frac{1}{2} - \frac{\sqrt{3}}{6}, \qquad c_2 = \frac{1}{2} + \frac{\sqrt{3}}{6}.$$

Integrating their Lagrange basis functions gives

$$\begin{array}{c|cc} \frac{1}{2} - \frac{\sqrt{3}}{6} & \frac{1}{4} & \frac{1}{4} - \frac{\sqrt{3}}{6} \\ \frac{1}{2} + \frac{\sqrt{3}}{6} & \frac{1}{4} + \frac{\sqrt{3}}{6} & \frac{1}{4} \\ \hline & \frac{1}{2} & \frac{1}{2} \end{array}. \tag{117}$$

In particular, the upper-right entry has a minus sign, so that the first row sums to c_1. The stage equations are

$$k_1 = f\left(t_n + c_1\tau, x_n + \tau\left[\frac{1}{4}k_1 + \left(\frac{1}{4} - \frac{\sqrt{3}}{6}\right)k_2\right]\right),$$
$$k_2 = f\left(t_n + c_2\tau, x_n + \tau\left[\left(\frac{1}{4} + \frac{\sqrt{3}}{6}\right)k_1 + \frac{1}{4}k_2\right]\right),$$
$$x_{n+1} = x_n + \frac{\tau}{2}(k_1 + k_2).$$

Both slopes must be solved for together. The method has order four, and substitution into (101) gives

$$R_{\mathrm{G},2}(z) = \frac{1 + z/2 + z^2/12}{1 - z/2 + z^2/12}. \tag{118}$$

13.4. **Quadratic first integrals.** For the autonomous equation $x' = f(x)$, a function $I(x)$ is a *first integral* if it is constant along every solution. Consider a quadratic function

$$I(x) = x^{\mathsf{T}}Qx, \qquad Q^{\mathsf{T}} = Q.$$

Positive definiteness is not needed for conservation. Since

$$\frac{d}{dt}I(x(t)) = 2x(t)^{\mathsf{T}}Qf(x(t)),$$

the condition for a quadratic first integral is

$$x^{\mathsf{T}}Qf(x) = 0 \qquad \text{for every } x. \tag{119}$$

Theorem 8 (Quadratic invariants of Gauss methods)**.** *Every Gauss collocation method preserves each quadratic first integral exactly, provided its stage equations are solved exactly.*

Proof. Let $U(\theta) = u(t_n + \tau\theta)$ and $U_j = U(c_j)$. The polynomial U has degree at most s, and

$$U'(c_j) = \tau f(U_j).$$

Therefore $2U^{\mathsf{T}}QU'$ has degree at most $2s-1$ and Gaussian quadrature is exact for it. Consequently,

$$I(x_{n+1}) - I(x_n) = \int_0^1 2U(\theta)^{\mathsf{T}}QU'(\theta)\,d\theta$$

$$= 2\tau \sum_{j=1}^{s} b_j U_j^{\mathsf{T}} Q f(U_j) = 0.$$

The last equality uses (119) at each stage value. □

For example, if $f(x) = Ax$ and $A^{\mathsf{T}}Q + QA = 0$, the quadratic form is preserved by the exact solution and by every Gauss method. Taking $Q = I$ and $A^{\mathsf{T}} = -A$ recovers norm preservation for the oscillator from September 18. The argument relies on the invariant being quadratic; it does not give exact preservation of a general nonlinear first integral.

13.5. **An energy identity for the scalar test equation.** Apply Gauss collocation to $x' = \lambda x$, with $z = \tau\lambda$. Now U is complex-valued and $U'(c_j) = zU_j$. Since the derivative of $|U|^2$ is a real polynomial of degree at most $2s - 1$, exact quadrature gives

$$\begin{aligned} |x_{n+1}|^2 - |x_n|^2 &= 2\int_0^1 \operatorname{Re}\big(\overline{U(\theta)}U'(\theta)\big)\, d\theta \\ &= 2\operatorname{Re} z \sum_{j=1}^{s} b_j |U_j|^2. \end{aligned} \tag{120}$$

For $\operatorname{Re} z < 0$, positivity of the weights implies $|x_{n+1}| \le |x_n|$.

This also proves unique solvability in the open left half-plane. Indeed, for a homogeneous stage system take $x_n = 0$. The left side is then nonnegative and the right side nonpositive, so every $U_j = 0$. Since $U(0) = 0$ and the Gauss nodes are distinct and interior, U has $s+1$ distinct zeros and must vanish identically. Thus the stage matrix has trivial kernel.

For $x_n \neq 0$, the sum in (120) is strictly positive: if all $U_j = 0$, then all $U'(c_j) = 0$, so $U' \equiv 0$ and $U \equiv 0$, a contradiction. Therefore

$$\operatorname{Re} z < 0 \quad \Longrightarrow \quad |R(z)| < 1.$$

Every Gauss RK method is consequently A-stable. On the imaginary axis, the same identity gives $|R(z)| = 1$ whenever the stages are defined, explaining the absence of artificial damping for oscillatory modes. We continue this comparison with Radau methods in the next lecture.

14. September 25, 2026

We compare Gauss and Radau collocation through their effects on quadratic quantities. Gauss quadrature gives an exact energy identity; right Radau quadrature contributes an additional nonpositive term. This leads to stability and contractivity results, and explains why Radau methods can damp very stiff modes more effectively.

As before, $\tau = \tau_n$, $\mathsf{A} = (a_{ij})$, and $z = \tau\lambda$. We use the normalized collocation polynomial

$$U(\theta) = u(t_n + \tau\theta), \qquad U_j = U(c_j), \qquad 0 \le \theta \le 1.$$

Its derivative is with respect to θ, so $U'(c_j) = \tau f(t_n + c_j\tau, U_j)$.

14.1. **Gauss stability revisited.** On the test equation $x' = \lambda x$, the numerical update is $x_{n+1} = R(z)x_n$, and

$$|x_{n+1}|^2 - |x_n|^2 = (|R(z)|^2 - 1)|x_n|^2.$$

Thus absolute stability is equivalent to nonincrease of the squared modulus. For Gauss collocation, the energy identity from (120) is

$$|x_{n+1}|^2 - |x_n|^2 = 2\,\mathrm{Re}\, z \sum_{j=1}^{s} b_j |U_j|^2. \tag{121}$$

There is no quadrature remainder: the integrand $2\,\mathrm{Re}(\overline{U}U')$ has degree at most $2s-1$, exactly the Gaussian degree of precision.

For nonzero initial data and a uniquely solvable stage system, the sum on the right is positive. Hence

$$\begin{array}{l|l} \mathrm{Re}\, z < 0 & |R(z)| < 1 \\ \mathrm{Re}\, z = 0 & |R(z)| = 1 \\ \mathrm{Re}\, z > 0 & |R(z)| > 1 \end{array}.$$

As proved on September 23, the stages are uniquely solvable throughout the open left half-plane. In the notation used throughout these notes, the strict region $\mathcal{S}$ is the open left half-plane, and the non-strict domain $\mathcal{D}$ also includes admissible points on the imaginary axis.

14.2. **Right Radau quadrature.** Right Radau quadrature fixes the last node at the endpoint,

$$0 < c_1 < \cdots < c_{s-1} < c_s = 1.$$

For $s \geq 2$, its interior nodes are the roots of the shifted Jacobi polynomial $P_{s-1}^{(1,0)}(2\theta - 1)$. Equivalently, the monic polynomial

$$\pi_{s-1}(\theta) = \prod_{j=1}^{s-1}(\theta - c_j)$$

is orthogonal to $\mathcal{P}_{s-2}$ with weight $1 - \theta$:

$$\int_0^1 (1-\theta)\pi_{s-1}(\theta)q(\theta)\,d\theta = 0, \qquad q \in \mathcal{P}_{s-2}.$$

For $s = 1$, take $\pi_0 = 1$ and the sole node $c_1 = 1$. The interpolatory weights are again $b_j = \int_0^1 \ell_j(\theta)\,d\theta$.

The degree of precision is $2s - 2$. Indeed, divide a polynomial of degree at most $2s - 2$ by $(\theta - 1)\pi_{s-1}(\theta)$. The remainder has degree at most $s - 1$, so it is integrated exactly by interpolation. The other term has zero integral by the weighted orthogonality and vanishes at every node. As for Gauss quadrature,

$$b_j = \int_0^1 \ell_j(\theta)^2\,d\theta > 0,$$

because $\deg \ell_j^2 \leq 2s - 2$.

Collocation at these nodes gives the *Radau IIA* family. The endpoint choice $c_s = 1$ specifies the right Radau family used in this lecture. Its order is

$$(2s - 2) + 1 = 2s - 1.$$

14.3. **The sign of the Radau quadrature error.** On $[0, 1]$, let

$$E[\varphi] = \int_0^1 \varphi(\theta)\,d\theta - \sum_{j=1}^{s} b_j\varphi(c_j).$$

Define the positive constant

$$\kappa_s = \int_0^1 (1-\theta)\pi_{s-1}(\theta)^2\,d\theta = \frac{s\big((s-1)!\big)^4}{2\big((2s-1)!\big)^2} > 0. \tag{122}$$

The polynomial $(\theta-1)\pi_{s-1}(\theta)^2$ is monic of degree $2s-1$ and vanishes at every quadrature node. Its lower degree terms are integrated exactly, so

$$E[\theta^{2s-1}] = -\kappa_s. \tag{123}$$

In particular, precision cannot exceed $2s-2$.

For a real function $\varphi \in C^{2s-1}([0,1])$, the usual remainder formula is

$$E[\varphi] = -\frac{\kappa_s}{(2s-1)!}\varphi^{(2s-1)}(\xi), \qquad \xi \in [0,1]. \tag{124}$$

One way to obtain it is to use the Hermite interpolant matching φ and φ' at each interior node and matching φ at 1. Its remainder contains the factor $(\theta-1)\pi_{s-1}(\theta)^2$, which has one sign on the interval. Integration then gives (124). For the energy argument below, the polynomial identity (123) is already sufficient.

14.4. **The Radau energy identity and A-stability.** Write the complex scalar collocation polynomial as

$$U(\theta) = \alpha_s\theta^s + \cdots + \alpha_0.$$

The highest-degree term in the derivative of $|U|^2$ is

$$\frac{d}{d\theta}|U(\theta)|^2 = 2s|\alpha_s|^2\theta^{2s-1} + \text{terms of degree at most } 2s-2.$$

Consequently its Radau quadrature error is $-2s\kappa_s|\alpha_s|^2$. On $x' = \lambda x$, $U'(c_j) = zU_j$, so

$$\boxed{|x_{n+1}|^2 - |x_n|^2 = 2\,\mathrm{Re}\, z \sum_{j=1}^{s} b_j|U_j|^2 - 2s\kappa_s|\alpha_s|^2.} \tag{125}$$

Both terms on the right are nonpositive when $\mathrm{Re}\, z \le 0$. Thus $|R(z)| \le 1$, provided the stage system is solvable.

The identity also proves solvability in the closed left half-plane. For a homogeneous stage system, $U(0) = 0$. Equation (125) forces $U(1) = 0$ and $\alpha_s = 0$. If $\mathrm{Re}\, z < 0$, it also forces all stage values to vanish, so $U \equiv 0$. If $\mathrm{Re}\, z = 0$, the polynomial $U' - zU$ has degree at most $s-1$ and vanishes at all s nodes. It is therefore identically zero. Together

with $U(0) = 0$, this again gives $U \equiv 0$. Hence the stage matrix has trivial kernel. Every Radau IIA method is A-stable.

14.5. **Quadratic energy.** For a real autonomous system, let

$$I(x) = x^{\mathsf{T}} Q x, \qquad Q^{\mathsf{T}} = Q.$$

The exact derivative is

$$\frac{d}{dt} I(x(t)) = 2x(t)^{\mathsf{T}} Q f(x(t)).$$

Thus $x^{\mathsf{T}} Q f(x) = 0$ gives a first integral, while $x^{\mathsf{T}} Q f(x) \le 0$ gives a nonincreasing quadratic quantity. Write $U(\theta) = \alpha_s \theta^s + \cdots + \alpha_0$, now with vector coefficients. For Gauss and Radau IIA, respectively, the quadrature arguments give

$$I(x_{n+1}) - I(x_n) = 2\tau \sum_{j=1}^{s} b_j U_j^{\mathsf{T}} Q f(U_j), \tag{126}$$

$$I(x_{n+1}) - I(x_n) = 2\tau \sum_{j=1}^{s} b_j U_j^{\mathsf{T}} Q f(U_j) - 2s\kappa_s \alpha_s^{\mathsf{T}} Q \alpha_s. \tag{127}$$

For Gauss methods, a quadratic first integral is preserved for any symmetric Q. For Radau methods, the last term is nonpositive when $Q \succeq 0$; it generally causes numerical dissipation even when the exact quantity is conserved. More generally, if

$$Q \succeq 0, \qquad x^{\mathsf{T}} Q f(x) \le 0 \quad \text{for all } x,$$

then both families satisfy

$$x_{n+1}^{\mathsf{T}} Q x_{n+1} \le x_n^{\mathsf{T}} Q x_n.$$

The assumption $Q \succeq 0$ matters for the Radau conclusion: an indefinite quadratic form does not give a fixed sign for the remainder term.

14.6. **One-stage and two-stage Radau IIA methods.** For $s = 1$, the tableau is

$$\begin{array}{c|c} 1 & 1 \\ \hline & 1 \end{array},$$

so the method is backward Euler, of order one.

For $s = 2$, the nodes and weights are

$$c_1 = \frac{1}{3}, \qquad c_2 = 1, \qquad b_1 = \frac{3}{4}, \qquad b_2 = \frac{1}{4}.$$

The Lagrange basis is

$$\ell_1(\theta) = \frac{3}{2}(1-\theta), \qquad \ell_2(\theta) = \frac{3}{2}\theta - \frac{1}{2}.$$

Integrating up to each node gives the tableau

$$\begin{array}{c|cc} \frac{1}{3} & \frac{5}{12} & -\frac{1}{12} \\ 1 & \frac{3}{4} & \frac{1}{4} \\ \hline & \frac{3}{4} & \frac{1}{4} \end{array}. \tag{128}$$

Hence

$$\begin{aligned} k_1 &= f\left(t_n + \frac{\tau}{3}, x_n + \tau\left[\frac{5}{12}k_1 - \frac{1}{12}k_2\right]\right), \\ k_2 &= f\left(t_n + \tau, x_n + \tau\left[\frac{3}{4}k_1 + \frac{1}{4}k_2\right]\right), \\ x_{n+1} &= x_n + \tau\left(\frac{3}{4}k_1 + \frac{1}{4}k_2\right). \end{aligned}$$

The quadrature precision is two, so the method has order three. For example, the moments through degree two are

$$\frac{3}{4} + \frac{1}{4} = 1, \qquad \frac{3}{4}\frac{1}{3} + \frac{1}{4} = \frac{1}{2}, \qquad \frac{3}{4}\frac{1}{9} + \frac{1}{4} = \frac{1}{3}.$$

Its stability function is

$$R_{\mathrm{R},2}(z) = \frac{1 + z/3}{1 - 2z/3 + z^2/6}. \tag{129}$$

14.7. **Dissipative ODEs and contractivity.**

Definition 12 (Dissipative vector field)**.** *The autonomous ODE $x' = f(x)$ is* dissipative *in the Euclidean inner product if*

$$\langle f(x) - f(y), x - y\rangle \le 0 \qquad \textit{for all } x, y. \tag{130}$$

For two exact solutions,

$$\frac{d}{dt}\|x(t) - y(t)\|_2^2 = 2\langle f(x(t)) - f(y(t)), x(t) - y(t)\rangle \le 0.$$

Therefore

$$\|x(t) - y(t)\|_2 \le \|x(t_0) - y(t_0)\|_2, \qquad t \ge t_0. \tag{131}$$

This concerns the distance between any two solutions. If $f(0) = 0$, taking $y(t) = 0$ also gives nonincrease of $\|x(t)\|_2$.

Gauss and Radau IIA preserve the same contractivity property. To see this, let U, V be the collocation polynomials for starting values x_n, y_n on the same step, and put $W = U - V$. At the nodes,

$$W'(c_j) = \tau\big(f(U_j) - f(V_j)\big).$$

The Gauss energy identity applied to W gives

$$\|x_{n+1} - y_{n+1}\|_2^2 - \|x_n - y_n\|_2^2 = 2\tau \sum_{j=1}^{s} b_j \langle U_j - V_j, f(U_j) - f(V_j)\rangle \le 0.$$

For Radau IIA, an additional term $-2s\kappa_s\|\alpha_s(U) - \alpha_s(V)\|_2^2$ appears, which is also nonpositive. Thus both methods satisfy

$$\boxed{\|x_{n+1} - y_{n+1}\|_2 \le \|x_n - y_n\|_2.} \tag{132}$$

The conclusion holds whenever the stage equations have been solved. It imposes no additional step-size restriction for contractivity. This preservation of contractivity for dissipative nonlinear ODEs is commonly called *B-stability*.

14.8. **Stiff accuracy and L-stability.** An A-stable method is L-stable if

$$R(z) \longrightarrow 0 \quad \text{as } |z| \to \infty \text{ in the left half-plane.}$$

This additional condition suppresses modes whose decay times are much shorter than the chosen step.

Suppose A is invertible. For large $|z|$,

$$\begin{aligned}(I - z\mathsf{A})^{-1} &= -z^{-1}(\mathsf{A} - z^{-1}I)^{-1} \\ &= -z^{-1}\mathsf{A}^{-1} - z^{-2}\mathsf{A}^{-2} + O(|z|^{-3}).\end{aligned}$$

Substitution into the RK stability function gives

$$R(z) = 1 - b^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{1} + O(|z|^{-1}). \tag{133}$$

Thus, for an A-stable method with invertible A, the additional condition for L-stability is

$$b^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{1} = 1. \tag{134}$$

For Radau IIA, $c_s = 1$, so the coefficient formula gives

$$a_{sj} = \int_0^{c_s} \ell_j(\theta)\, d\theta = \int_0^1 \ell_j(\theta)\, d\theta = b_j.$$

The last row of A therefore equals b^{T}. This property, together with $c_s = 1$, is called *stiff accuracy*: the last stage value is the new solution,

$$U_s = x_n + \tau \sum_j a_{sj} k_j = x_{n+1}.$$

If e_s is the last coordinate vector, then

$$b^{\mathsf{T}} = e_s^{\mathsf{T}}\mathsf{A} \quad \Longrightarrow \quad b^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{1} = e_s^{\mathsf{T}}\mathbf{1} = 1.$$

For completeness, A is invertible for collocation at distinct nonzero nodes. If $\mathsf{A}v = 0$, the polynomial

$$V(\theta) = \int_0^{\theta} \sum_{j=1}^{s} v_j \ell_j(\xi)\, d\xi$$

has degree at most s and vanishes at $0, c_1, \ldots, c_s$. Hence $V \equiv 0$, so $v_j = V'(c_j) = 0$ for every j. This applies to both Gauss and Radau IIA nodes.

Combining invertibility, stiff accuracy, and A-stability proves that every Radau IIA method is L-stable. The endpoint condition $c_s = 1$ yields the row identity for collocation methods; it would not by itself imply stiff accuracy for an arbitrary RK tableau.

Gauss methods are not L-stable. Their invertible coefficient matrices ensure a finite limit in (133), while their unit modulus on the imaginary axis forces that limit to have modulus one. For the first two methods,

the difference is explicit:

method	$R(z)$	$\lim_{r\to\infty} R(-r)$
Gauss, $s=1$	$\dfrac{1+z/2}{1-z/2}$	-1
Gauss, $s=2$	$\dfrac{1+z/2+z^2/12}{1-z/2+z^2/12}$	1
Radau IIA, $s=1$	$\dfrac{1}{1-z}$	0
Radau IIA, $s=2$	$\dfrac{1+z/3}{1-2z/3+z^2/6}$	0

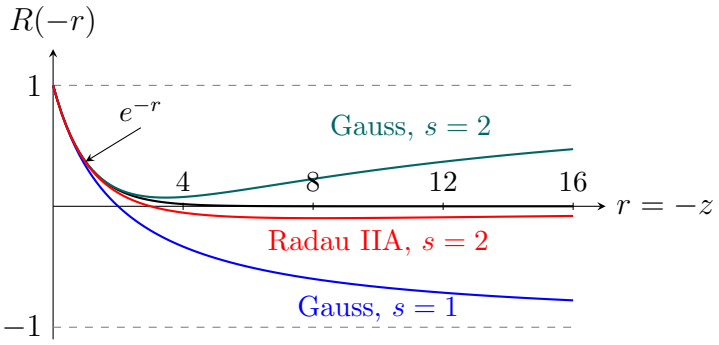


FIGURE 12. Amplification of a decaying scalar mode. All three numerical methods are A-stable, but the Radau factor tends to zero for very stiff modes. The Gauss factors eventually approach -1 and 1, respectively.

15. SEPTEMBER 28, 2026

We revisit stability of Gauss and Radau methods, interpret their stability functions as Padé approximations to the exponential, and study reversibility under a change in the direction of time. We retain A for the matrix in a linear ODE, $\mathsf{A} = (a_{ij})$ for the RK coefficient matrix, and $\tau = t_{n+1} - t_n$. As in the earlier lectures, Φ is the exact flow and Ψ is the numerical flow.

15.1. **Linear stability and stiff decay.** For the constant-coefficient system $x' = Ax$, the exact and RK updates are

$$x(t_{n+1}) = e^{\tau A} x(t_n), \qquad x_{n+1} = R(\tau A) x_n.$$

Here R is the rational stability function

$$R(z) = 1 + zb^{\mathsf{T}}(I - z\mathsf{A})^{-1}\mathbf{1}, \qquad z = \tau\lambda. \tag{135}$$

If $R = P/Q$, its matrix evaluation means $R(\tau A) = Q(\tau A)^{-1}P(\tau A)$, whenever defined; the RK stage system must also be solvable.

Recall the distinction between the strict stability region $\mathcal{S}$, where $|R(z)| < 1$, and the non-strict domain $\mathcal{D}$, where $|R(z)| \leq 1$. In both cases we require the stages to be defined. The handwritten notes use the non-strict convention in this lecture. An A-stable method has

$$\{z \in \mathbb{C} : \operatorname{Re} z \leq 0\} \subseteq \mathcal{D}.$$

Gauss and Radau IIA methods are A-stable for every number of stages $s \geq 1$.

For the differential equation, asymptotic stability of the origin means that every solution tends to zero, and is equivalent to

$$\operatorname{Re}\lambda < 0 \qquad \text{for all } \lambda \in \sigma(A). \tag{136}$$

The sign is negative. Mere boundedness allows eigenvalues on the imaginary axis, provided they are semisimple, as discussed on September 11. Likewise, at a fixed positive step size,

$$R(\tau A)^n \longrightarrow 0 \quad \Longleftrightarrow \quad |R(\tau\lambda)| < 1 \quad \text{for all } \lambda \in \sigma(A).$$

Gauss and Radau IIA satisfy this strict inequality in the open left half-plane, so both reproduce asymptotic decay for every $\tau > 0$ when (136) holds.

Definition 13 (L-stability)**.** *An A-stable method is L-*stable *if*

$$R(z) \longrightarrow 0 \quad \textit{as } |z| \to \infty \textit{ in the left half-plane.}$$

In particular, for a fixed λ with $\operatorname{Re}\lambda < 0$, an L-stable method satisfies $R(\tau\lambda) \to 0$ as $\tau \to \infty$, matching the limit of $e^{\tau\lambda}$. This is a one-step damping property for very stiff modes, in addition to decay over many fixed steps. Radau IIA methods are L-stable; Gauss methods are not.

15.2. **Padé approximation of the exponential.**

Definition 14 (Padé approximant)**.** *The $[m/n]$ Padé approximant of e^z is a rational function $P_m(z)/Q_n(z)$, with*

$$\deg P_m \le m, \qquad \deg Q_n \le n, \qquad Q_n(0) = 1,$$

whose Taylor expansion agrees with that of e^z through degree $m+n$:

$$Q_n(z)e^z - P_m(z) = O(z^{m+n+1}). \tag{137}$$

An RK method of order p, applied to the scalar test equation, necessarily satisfies

$$R(z) = e^z + O(z^{p+1}).$$

This explains the connection between RK stability functions and rational approximations to the exponential. The Padé property requires the particular degree and matching conditions above; it does not hold for every RK stability function.

For an s-stage RK method, the determinant identity gives

$$R(z) = \frac{\det(I - z\mathsf{A} + z\mathbf{1}b^{\mathsf{T}})}{\det(I - z\mathsf{A})}.$$

Thus the numerator and denominator have degrees at most s. A Gauss method has order $2s$, so its stability function is the diagonal $[s/s]$ Padé approximant of e^z. The first two examples are

$$R_{\mathrm{G},1}(z) = \frac{1 + z/2}{1 - z/2}, \qquad R_{\mathrm{G},1}(z) - e^z = O(z^3),$$

$$R_{\mathrm{G},2}(z) = \frac{1 + z/2 + z^2/12}{1 - z/2 + z^2/12}, \qquad R_{\mathrm{G},2}(z) - e^z = O(z^5).$$

Consequently, on a linear system these methods give rational matrix approximations

$$e^{\tau A} \approx R(\tau A).$$

For example, the one-stage Gauss method replaces the exponential by $(I - \tau A/2)^{-1}(I + \tau A/2)$.

The diagonal exponential approximant can be written as

$$R_{\mathrm{G},s}(z) = \frac{P_s(z)}{P_s(-z)}, \qquad P_s(z) = \sum_{k=0}^{s} \frac{(2s-k)!\, s!}{(2s)!\, (s-k)!\, k!}\, z^k. \tag{138}$$

Its numerator and denominator have the same degree, and their leading coefficients differ by $(-1)^s$. Hence

$$\lim_{|z|\to\infty} R_{\mathrm{G},s}(z) = (-1)^s,$$

which explains the failure of L-stability.

15.2.1. *The nonlinear flow operator.* For an autonomous equation $x' = f(x)$ in $\mathbb{R}^d$, introduce the differential operator

$$\mathcal{L}g(x) = \sum_{j=1}^{d} f_j(x)\frac{\partial g}{\partial x_j}(x) = Dg(x)f(x).$$

Along the exact solution,

$$\frac{d}{dt}g(x(t)) = (\mathcal{L}g)(x(t)).$$

Thus the action of the exact flow on functions is formally

$$g(\Phi^\tau(x)) = (e^{\tau\mathcal{L}}g)(x) = g(x) + \tau\mathcal{L}g(x) + \frac{\tau^2}{2}\mathcal{L}^2 g(x) + \cdots.$$

For smooth functions this can be read as a finite Taylor expansion with a remainder; convergence of the infinite series needs further assumptions. This is the operator analogy with the matrix exponential mentioned in class. For general nonlinear f, the RK map is not in general obtained by simply substituting $\tau\mathcal{L}$ into its scalar stability function. The formula $R(\tau A)x$ is exact for the numerical update of a linear system.

15.3. **Why Radau IIA is L-stable.** Suppose A is invertible. From (135),

$$z(I - z\mathsf{A})^{-1} = -\mathsf{A}^{-1} + O(|z|^{-1}),$$

and therefore

$$\lim_{|z|\to\infty} R(z) = 1 - b^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{1}. \tag{139}$$

If b^{T} coincides with the last row of A, then

$$b^{\mathsf{T}} = e_s^{\mathsf{T}}\mathsf{A} \quad\Longrightarrow\quad 1 - b^{\mathsf{T}}\mathsf{A}^{-1}\mathbf{1} = 1 - e_s^{\mathsf{T}}\mathbf{1} = 0.$$

Here e_s is the last coordinate vector. Together with A-stability, this proves L-stability.

For collocation, the coefficients are

$$b_j = \int_0^1 \ell_j(\theta)\, d\theta, \qquad a_{ij} = \int_0^{c_i} \ell_j(\theta)\, d\theta.$$

Right Radau nodes include $c_s = 1$, so $a_{sj} = b_j$. This is the stiff accuracy property from September 25, and the last stage equals x_{n+1}. The nodes are distinct and nonzero, so A is invertible by the polynomial argument given there. Since Radau IIA is also A-stable, every member of the family is L-stable.

For the first two members,

$$R_{\mathrm{R},1}(z) = \frac{1}{1-z}, \qquad R_{\mathrm{R},2}(z) = \frac{1 + z/3}{1 - 2z/3 + z^2/6}.$$

Both tend to zero as $|z| \to \infty$. More generally, the denominator has degree s, and the zero limit forces the numerator to have degree at most $s-1$. Since Radau IIA has order $2s-1$, its stability function is the $[s-1/s]$ Padé approximant of the exponential.

15.4. **Time reversibility of a one-step method.** For the exact flow of $x' = f(t,x)$, uniqueness implies

$$\Phi^{t_2,t_1}(x(t_1)) = x(t_2), \qquad \Phi^{t_1,t_2}(x(t_2)) = x(t_1).$$

Whenever both directions of the flow exist,

$$\Phi^{t_1,t_2} \circ \Phi^{t_2,t_1} = I. \tag{140}$$

Advancing along a solution and then returning along the same solution recovers the initial value.

Definition 15. *A one-step method is* time-reversible*, or* symmetric*, if*

$$\Psi^{t_n,t_{n+1}} \circ \Psi^{t_{n+1},t_n} = I. \tag{141}$$

The backward map uses the same method with step $-\tau$. For an autonomous equation this reads $\Psi^{-\tau} \circ \Psi^{\tau} = I$.

For implicit methods, this statement is understood where the forward and backward steps are well defined, using the locally unique solution branches of their stage equations.

Example 19 (Forward Euler is not reversible)**.** *A forward step gives*

$$x_{n+1} = x_n + \tau f(t_n, x_n).$$

Applying forward Euler with step $-\tau$ *from* t_{n+1} *gives a returned value*

$$\begin{aligned}\widehat{x}_n &= x_{n+1} - \tau f(t_{n+1}, x_{n+1}) \\ &= x_n + \tau f(t_n, x_n) - \tau f(t_{n+1}, x_{n+1}),\end{aligned}$$

which generally differs from x_n*. On* $x' = \lambda x$*, the forward and backward multipliers give*

$$\widehat{x}_n = (1 - z)(1 + z)x_n = (1 - z^2)x_n.$$

15.5. **Reversibility of Gauss collocation.**

Theorem 9. *Gauss collocation methods are time-reversible wherever the relevant collocation solutions are uniquely defined.*

Proof. Let $t_{n+1} = t_n + \tau$, and let $u \in (\mathcal{P}_s)^d$ be the forward collocation polynomial:

$$u(t_n) = x_n, \qquad u'(t_n + c_i\tau) = f(t_n + c_i\tau, u(t_n + c_i\tau)),$$

with $x_{n+1} = u(t_{n+1})$. The backward step seeks $v \in (\mathcal{P}_s)^d$ satisfying

$$v(t_{n+1}) = x_{n+1}, \qquad v'(t_{n+1} - c_i\tau) = f(t_{n+1} - c_i\tau, v(t_{n+1} - c_i\tau)).$$

Gauss nodes obey $c_{s+1-i} = 1 - c_i$. Consequently,

$$t_{n+1} - c_i\tau = t_n + (1 - c_i)\tau = t_n + c_{s+1-i}\tau.$$

The backward collocation points are precisely the forward collocation points in reverse order. The polynomial u therefore satisfies all the backward conditions as well. Uniqueness gives $v = u$, and the returned value is $v(t_n) = u(t_n) = x_n$, proving (141). □

The derivative in this proof is taken with respect to physical time; the backward step changes the node locations through $-\tau$. Equivalently,

in the normalized variable, $V(\theta) = U(1-\theta)$ has $V'(\theta) = -U'(1-\theta)$, which supplies the negative step in the backward collocation equations.

On the scalar test equation, reversibility requires

$$R(-z)R(z) = 1$$

where both steps are defined. The Gauss Padé formula (138) has exactly this property. For the one-stage method, it is also immediate from

$$x_{n+1} = x_n + \tau f\left(t_n + \frac{\tau}{2}, \frac{x_n + x_{n+1}}{2}\right):$$

interchanging the endpoints and replacing τ by $-\tau$ leaves the equation unchanged. The collocation proof extends this symmetry to every Gauss method and to nonlinear, time-dependent vector fields.

16. September 30, 2026

We continue the discussion of time reversibility from September 28. Symmetry constrains the error expansion and makes extrapolation especially effective. We also introduce variable doubling and the perturbation viewpoint behind modified equations. Throughout, Φ and Ψ denote the exact and numerical flows, and τ denotes a constant step size.

16.1. **Reversible integrators and explicit RK methods.** Recall that a one-step method is symmetric if

$$\Psi^{t,t+\tau} \circ \Psi^{t+\tau,t} = I,$$

where the first map uses the same method with step $-\tau$. Gauss collocation has this property because its nodes are invariant under reflection about the midpoint of the step.

Proposition 2. *A consistent explicit RK method with finitely many stages cannot be symmetric for general ODEs.*

Proof. Apply the method to $x' = x$, with initial value one. Its stability function $R(z)$ is a polynomial, and consistency gives $R(0) = R'(0) = 1$. A forward step followed by a backward step multiplies the initial value

by $R(-\tau)R(\tau)$. Symmetry would imply

$$R(-z)R(z) = 1$$

for all sufficiently small real z, hence as a polynomial identity. If R has degree $m \geq 1$ and leading coefficient a_m, the product has degree $2m$ and leading coefficient $(-1)^m a_m^2 \neq 0$. It cannot equal the constant one. □

This obstruction concerns standard explicit RK methods on a general vector field. A method exploiting a partition of the variables can still be both explicit and reversible.

16.2. **Variable doubling.** For $x' = f(t, x)$, introduce a second copy of the unknown and consider

$$x' = f(t, y), \qquad y' = f(t, x), \qquad x(t_0) = y(t_0) = x_0. \tag{142}$$

The diagonal $x = y$ is invariant: the pair consisting of two copies of the original solution solves this system, and uniqueness identifies it with the solution of (142).

The following staggered update is explicit:

$$\begin{aligned} y_{n+1/2} &= y_n + \frac{\tau}{2} f(t_n, x_n), \\ x_{n+1} &= x_n + \tau f(t_n + \tau/2, y_{n+1/2}), \\ y_{n+1} &= y_{n+1/2} + \frac{\tau}{2} f(t_n + \tau, x_{n+1}). \end{aligned} \tag{143}$$

Equivalently,

$$\begin{aligned} x_{n+1} &= x_n + \tau f\left(t_n + \frac{\tau}{2}, y_n + \frac{\tau}{2} f(t_n, x_n)\right), \\ y_{n+1} &= y_n + \frac{\tau}{2}\big(f(t_n, x_n) + f(t_n + \tau, x_{n+1})\big). \end{aligned}$$

Here $y_{n+1/2}$ is an intermediate variable, not an additional solution value required from a previous step.

To reverse the step, start from (x_{n+1}, y_{n+1}) at $t_n + \tau$ and replace τ by $-\tau$. The three operations successively recover

$$y_{n+1} - \frac{\tau}{2} f(t_n + \tau, x_{n+1}) = y_{n+1/2}, \qquad x_n, \qquad y_n.$$

Thus the map on the pair (x, y) is reversible.

Taylor expansion also shows that it has order two for the doubled system. For example, its first component satisfies

$$x_{n+1} = x_n + \tau f(t_n, y_n) + \frac{\tau^2}{2}\big(f_t(t_n, y_n) + f_x(t_n, y_n) f(t_n, x_n)\big) + O(\tau^3),$$

which agrees with the exact expansion of $x' = f(t, y)$. The corresponding statement holds for the second component. Numerically, x_n and y_n need not be equal. Both copies must be retained between steps for the reversibility argument to apply; resetting one copy to the other defines a different method.

16.3. **Even powers in the global error expansion.** Write $x_\tau(t_n) = x_n$ for the numerical solution computed with constant step τ and exact initial data. At a fixed endpoint, $x_\tau(T) = Y_\tau(T)$ in the notation of September 9. Comparisons between step sizes use common grid points on a fixed finite interval.

Theorem 10 (Error expansion for symmetric methods)**.** *Suppose a smooth, consistent symmetric one-step method has finite order and admits the usual asymptotic error expansion. Its order is even, say $2r$, and, with sufficient smoothness,*

$$x_\tau(t) = x(t) + C_{2r}(t)\tau^{2r} + C_{2r+2}(t)\tau^{2r+2} + \cdots . \tag{144}$$

The expansion is understood to any finite order for which the required derivatives exist; no convergence of the infinite series is asserted.

One explanation uses the formal modified equation developed next lecture. In the autonomous setting, write the numerical map formally as the time-τ flow of a step-dependent vector field g_τ. Symmetry says that its inverse is the same map with step $-\tau$. Taking the formal logarithm gives

$$-\tau g_{-\tau} = -\tau g_\tau, \qquad g_{-\tau} = g_\tau.$$

Thus only even powers of τ occur in g_τ, and hence in the fixed-time solution expansion. A nonautonomous equation can be treated by adjoining $t' = 1$. The statement requires the fixed step and initialization

assumptions above; arbitrary interpolation between grid points or arbitrary step changes do not automatically preserve this expansion.

16.4. **Richardson extrapolation in the squared step size.** We specialize (44) to a second-order symmetric method:

$$x_\tau(t) = x(t) + C_2(t)\tau^2 + C_4(t)\tau^4 + O(\tau^6),$$
$$x_{\tau/2}(t) = x(t) + \frac{1}{4}C_2(t)\tau^2 + \frac{1}{16}C_4(t)\tau^4 + O(\tau^6).$$

Eliminating the leading error gives

$$\boxed{\frac{4x_{\tau/2}(t) - x_\tau(t)}{3} = x(t) - \frac{1}{4}C_4(t)\tau^4 + O(\tau^6).} \tag{145}$$

One extrapolation therefore raises the order from two to four. For an expansion beginning at order $2r$, the corresponding combination is

$$\frac{2^{2r}x_{\tau/2}(t) - x_\tau(t)}{2^{2r} - 1} = x(t) + O(\tau^{2r+2}).$$

Using three step sizes, fit a quadratic polynomial in $s = \tau^2$:

$$P(s) = d_0 + d_2 s + d_4 s^2,$$

with values $x_\tau(t), x_{\tau/2}(t), x_{\tau/4}(t)$ at $s = \tau^2, \tau^2/4, \tau^2/16$. Componentwise, the system is

$$\begin{bmatrix} 1 & \tau^2 & \tau^4 \\ 1 & \tau^2/4 & \tau^4/16 \\ 1 & \tau^2/16 & \tau^4/256 \end{bmatrix} \begin{bmatrix} d_0 \\ d_2 \\ d_4 \end{bmatrix} = \begin{bmatrix} x_\tau(t) \\ x_{\tau/2}(t) \\ x_{\tau/4}(t) \end{bmatrix}.$$

Evaluating the interpolant at zero gives

$$\boxed{d_0 = \frac{1}{45}x_\tau(t) - \frac{20}{45}x_{\tau/2}(t) + \frac{64}{45}x_{\tau/4}(t) = x(t) + O(\tau^6).} \tag{146}$$

The weights sum to one and annihilate the τ^2 and τ^4 terms. This improves accuracy at the comparison time; a linear combination of numerical solutions need not retain the geometric properties of the underlying symmetric method.

16.5. **Perturbation expansions.** As a model for coefficient matching, consider weak damping:

$$q'' = -q - 2\varepsilon q', \qquad q(t) = q_0(t) + \varepsilon q_1(t) + \varepsilon^2 q_2(t) + \cdots .$$

Equating successive powers of ε gives

$$q_0'' = -q_0, \qquad q_1'' = -q_1 - 2q_0', \qquad q_m'' = -q_m - 2q_{m-1}' \quad (m \geq 1).$$

For initial data independent of ε, the higher corrections have zero initial position and velocity. For example, if $q(0) = 1$ and $q'(0) = 0$, then

$$q_0(t) = \cos t, \qquad q_1(t) = \sin t - t\cos t.$$

The secular term $t\cos t$ also reminds us that a fixed-time asymptotic expansion need not be uniform for arbitrarily long times.

For a numerical method, the small parameter is the step size. Forward error analysis expands its solution around the exact one; backward error analysis instead seeks a nearby differential equation whose flow agrees with the numerical method to a prescribed order. This leads to modified equations and, for gradient flows, an interpretation in terms of a modified potential.

17. October 2, 2026

We derive the leading global error of forward Euler and then construct its modified equation. These are two descriptions of the same leading perturbation: one changes the solution, while the other changes the vector field. We retain the convention that local error is exact minus numerical, whereas global error is numerical minus exact.

17.1. **Perturbing the starting value of a one-step map.** Consider $x' = f(t, x)$ and a smooth first-order consistent map

$$\Psi^{t+\tau,t}(x) = x + \tau f(t, x) + O(\tau^2).$$

Assume this expansion also holds after the needed derivatives in x, uniformly on the region under consideration. Taylor expansion with

respect to the starting value gives

$$\Psi^{t+\tau,t}(x+\delta x) = \Psi^{t+\tau,t}(x) + \delta x + \tau f_x(t,x)\delta x + O\big(\tau^2\|\delta x\| + |\tau|\|\delta x\|^2\big). \tag{147}$$

In particular, if $\delta x = O(\tau^m)$ with integer $m \geq 1$, the remainder is $O(\tau^{m+2})$. Here $f_x = D_x f$ is a Jacobian, so $f_x \delta x$ is a matrix–vector product. The identity term propagates an existing error, and the Jacobian describes its first-order change over the step.

17.2. **The local error of forward Euler.** Forward Euler is

$$x_{n+1} = x_n + \tau f(t_n, x_n). \tag{148}$$

Along an exact solution,

$$x''(t) = f_t(t,x(t)) + f_x(t,x(t))f(t,x(t)).$$

Consequently,

$$x(t+\tau) - \Psi^{t+\tau,t}(x(t)) = \tau^2 d_2(t) + O(\tau^3), \tag{149}$$

$$d_2(t) = \frac{1}{2}\big(f_t + f_x f\big)(t,x(t)) = \frac{1}{2}x''(t). \tag{150}$$

The subscript in d_2 records the power of τ. This is the leading local-error coefficient denoted by d_0 in the handwritten notes.

17.3. **An equation for the leading global error.** At grid points, seek an expansion

$$x_\tau(t) = x(t) + \tau C_1(t) + \tau^2 C_2(t) + \cdots, \qquad x_\tau(t_n) = x_n. \tag{151}$$

We use powers beginning at one, so C_1 is the coefficient called c_0 in the handwritten expansion. Exact initial data give $C_j(t_0) = 0$ for every correction.

Insert (151) into the Euler recurrence. The terms involving C_2 cancel at order τ^2, leaving

$$\tau^2 d_2(t) + \tau\big(C_1(t+\tau) - C_1(t)\big) - \tau^2 f_x(t,x(t))C_1(t) = O(\tau^3).$$

Divide by τ^2 and compare the leading terms. With $A(t) = f_x(t, x(t))$, we obtain

$$\boxed{C_1'(t) = A(t)C_1(t) - d_2(t), \qquad C_1(t_0) = 0.} \tag{152}$$

The minus sign agrees with the conventions above: a positive exact-minus-numerical local defect is subtracted in the recurrence for the numerical-minus-exact global error.

Let $W(t, s)$ be the fundamental matrix for the variational equation,

$$\partial_t W(t, s) = A(t)W(t, s), \qquad W(s, s) = I.$$

Variation of constants gives

$$C_1(t) = -\int_{t_0}^{t} W(t, s)d_2(s)\, ds. \tag{153}$$

Thus the leading local defects accumulate after being transported by the linearized exact flow. If the initial numerical value instead has expansion $x_0^{\text{num}} = x(t_0) + \tau\eta + O(\tau^2)$, the additional term is $W(t, t_0)\eta$.

Higher coefficients satisfy further inhomogeneous variational equations. Their forcing includes previously computed corrections, not just the next local-error coefficient. For example, for Euler, comparison at order τ^3 gives

$$C_2' = AC_2 + \frac{1}{2}f_{xx}[C_1, C_1] - \frac{1}{2}C_1'' - \frac{1}{6}x''', \qquad C_2(t_0) = 0,$$

where all derivatives of f are evaluated along $x(t)$.

Example 20 (Scalar linear equation)**.** *For $x' = \lambda x$, $d_2(t) = \lambda^2 x(t)/2$ and $W(t, s) = e^{\lambda(t-s)}$. Therefore*

$$C_1(t) = -\frac{\lambda^2}{2}(t - t_0)x(t).$$

This agrees with expanding $x_n = (1 + \tau\lambda)^n x_0$ at fixed $t_n - t_0 = n\tau$.

17.4. **Constructing a modified differential equation.** Backward error analysis seeks a step-dependent vector field

$$y' = g_\tau(t, y), \qquad g_\tau = f + \tau f_1 + \tau^2 f_2 + \cdots, \qquad y(t_0) = x_0, \tag{154}$$

whose exact time-τ map matches the numerical map order by order. This is a formal construction. A finite truncation generally approximates the numerical values at grid points; it does not interpolate them exactly for all step sizes.

To determine f_1 for Euler, expand one exact step of (154) from (t, y):

$$\begin{aligned} y(t+\tau) &= y + \tau g_\tau(t,y) + \frac{\tau^2}{2}\big(\partial_t g_\tau + D_y g_\tau\, g_\tau\big)(t,y) + O(\tau^3) \\ &= y + \tau f(t,y) + \tau^2\left[f_1(t,y) + \frac{1}{2}(f_t + f_x f)(t,y)\right] + O(\tau^3). \end{aligned}$$

Euler has no τ^2 term in its one-step map, so

(155)

$$\boxed{f_1 = -\frac{1}{2}(f_t + f_x f), \qquad y' = f(t,y) - \frac{\tau}{2}(f_t + f_x f)(t,y) + O(\tau^2).}$$

The factor $1/2$ comes from the second Taylor coefficient. For an autonomous equation this reduces to $y' = f(y) - \tau f'(y) f(y)/2 + O(\tau^2)$.

The connection with forward error analysis is direct. Expand the modified solution as $y = x + \tau y_1 + O(\tau^2)$. Then

$$y_1' = A(t) y_1 + f_1(t, x(t)), \qquad y_1(t_0) = 0.$$

Since $f_1(t, x(t)) = -d_2(t)$, this is precisely (152); hence $y_1 = C_1$.

If we retain only $f + \tau f_1$, its exact step and the Euler step differ by $O(\tau^3)$. Under uniform smoothness and stability bounds on a fixed finite interval, the resulting solution satisfies

$$x_n - x(t_n) = O(\tau), \qquad x_n - y(t_n) = O(\tau^2).$$

The same numerical values are therefore a more accurate approximation to the solution of this nearby equation.

For $x' = \lambda x$, the formal modified coefficient can even be read from the logarithm:

$$\frac{1}{\tau}\log(1+\tau\lambda) = \lambda - \frac{\tau}{2}\lambda^2 + \frac{\tau^2}{3}\lambda^3 - \cdots.$$

Near $\tau = 0$, its exponential over one step is exactly $1 + \tau\lambda$, consistent with (155).

17.5. **Gradient descent and a modified potential.** Let $V : \mathbb{R}^d \to \mathbb{R}$ be smooth and consider the gradient flow

$$x' = -\nabla V(x).$$

The potential decreases along exact trajectories because

$$\frac{d}{dt} V(x(t)) = -\|\nabla V(x(t))\|_2^2 \le 0.$$

Forward Euler is the gradient descent iteration

$$x_{n+1} = x_n - \tau \nabla V(x_n).$$

Here $f = -\nabla V$ and $f' = -\nabla^2 V$, so

$$f_1 = -\frac{1}{2} \nabla^2 V \, \nabla V.$$

Using $\nabla \|\nabla V\|_2^2 = 2 \nabla^2 V \, \nabla V$, the first modified equation becomes

$$\boxed{y' = -\nabla V_\tau(y), \qquad V_\tau = V + \frac{\tau}{4} \|\nabla V\|_2^2.} \tag{156}$$

Thus, to first order in the step size, discretization adds a penalty proportional to the squared gradient. This is the implicit regularization interpretation mentioned in class. The coefficient is $\tau/4$, as required by (155). This describes the truncated modified flow; it does not assert that Euler decreases either potential for an arbitrary step size.

18. October 5, 2026

We now turn from one-step methods to linear multistep methods. Instead of constructing new RK stages on every step, a multistep method reuses solution values and derivatives from earlier times. Throughout this discussion the grid is uniform:

$$t_n = t_0 + n\tau, \qquad \tau > 0, \qquad f_n = f(t_n, x_n).$$

The fixed coefficients below are for a constant step size.

18.1. **From one-step methods to methods with memory.** A one-step method computes x_{n+1} from x_n through $x_{n+1} = \Psi^{t_{n+1}, t_n}(x_n)$. A k-step method uses the history $x_n, \dots, x_{n+k-1}$ to determine x_{n+k}. It

therefore requires k starting values,

$$x_0, x_1, \dots, x_{k-1}.$$

Only x_0 is supplied by the IVP. The remaining values can be computed with a suitable one-step method, such as an RK method. Their accuracy will enter the multistep convergence estimate.

Example 21 (Explicit midpoint, or leapfrog)**.** *Integrating the ODE over two steps gives*

$$x(t_{n+2}) - x(t_n) = \int_{t_n}^{t_{n+2}} f(t, x(t))\, dt.$$

The midpoint quadrature rule suggests

$$\boxed{x_{n+2} - x_n = 2\tau f_{n+1}.} \tag{157}$$

This is an explicit two-step method. It retains both x_n and x_{n+1}, even though only one derivative appears in the formula. Equivalently, $x_{n+1} = x_{n-1} + 2\tau f_n$.

Example 22 (Simpson's multistep formula)**.** *Applying Simpson quadrature to the same integral gives*

$$\boxed{x_{n+2} - x_n = \frac{\tau}{3}\big(f_n + 4f_{n+1} + f_{n+2}\big).} \tag{158}$$

It is implicit because $f_{n+2} = f(t_{n+2}, x_{n+2})$ depends on the new value. It is still a two-step method: three time levels occur, but only two starting values are needed.

The multistep midpoint formula (157) is different from the one-step explicit midpoint RK method and from the implicit midpoint Gauss method. Their names refer to related quadrature ideas, but their update rules and stability properties are different.

18.2. **General form and characteristic polynomials.**

Definition 16 (Linear multistep method)**.** *A k-step LMM has the form*

$$\sum_{j=0}^{k} \alpha_j x_{n+j} = \tau \sum_{j=0}^{k} \beta_j f_{n+j}, \qquad \alpha_k \neq 0. \tag{159}$$

The coefficients are independent of n and τ. The method is explicit if $\beta_k = 0$, and implicit otherwise. We may normalize $\alpha_k = 1$ by dividing both sides by it.

The word "linear" refers to the linear combination of values and derivatives. An implicit LMM still requires a nonlinear solve when f is nonlinear. We assume that the representation has no unnecessary common factor in the two coefficient polynomials.

Define the shift operator on sequences by

$$Ex_n = x_{n+1}, \qquad E^j x_n = x_{n+j}, \qquad Ef_n = f_{n+1}.$$

The two characteristic polynomials are

$$\rho(\zeta) = \sum_{j=0}^{k} \alpha_j \zeta^j, \qquad \sigma(\zeta) = \sum_{j=0}^{k} \beta_j \zeta^j. \tag{160}$$

Then (159) can be written compactly as

$$\rho(E)x_n = \tau \sigma(E) f_n. \tag{161}$$

Here ζ is a polynomial variable; later, $z = \tau\lambda$ will denote the scaled eigenvalue in a stability test.

For the two examples,

method	$\rho(\zeta)$	$\sigma(\zeta)$
leapfrog	$\zeta^2 - 1$	2ζ
Simpson	$\zeta^2 - 1$	$\frac{1}{3}(\zeta^2 + 4\zeta + 1)$

This shift-operator form is also a finite linear filtering relation between the solution sequence and its derivative sequence.

18.3. **Consistency error and order.** Insert a smooth exact solution into the formula and define its unnormalized residual by

$$\mathcal{L}_\tau[x](t_n) = \sum_{j=0}^{k} \alpha_j x(t_n + j\tau) - \tau \sum_{j=0}^{k} \beta_j x'(t_n + j\tau). \tag{162}$$

This is $\rho(E)x(t_n) - \tau\sigma(E)x'(t_n)$, where the shift now acts on exact samples. It is the quantity denoted by L in the handwritten notes.

Definition 17 (Consistency order of an LMM)**.** *An LMM has order at least p if*

$$\mathcal{L}_\tau[x](t) = O(\tau^{p+1})$$

for every sufficiently smooth test function x. Its order is exactly p if the next coefficient does not vanish identically.

Some conventions divide the residual by τ, in which case an order-p method has normalized residual $O(\tau^p)$. We keep the unnormalized convention used in class. A residual is not itself a global error; propagation through the recurrence must also be controlled.

For leapfrog, expand about the midpoint t_{n+1}:

$$x(t_{n+2}) = x + \tau x' + \frac{\tau^2}{2}x'' + \frac{\tau^3}{6}x''' + \cdots,$$
$$x(t_n) = x - \tau x' + \frac{\tau^2}{2}x'' - \frac{\tau^3}{6}x''' + \cdots.$$

Subtracting and cancelling $2\tau x'$ gives

$$\mathcal{L}_\tau[x](t_n) = \frac{\tau^3}{3}x'''(t_{n+1}) + O(\tau^5). \tag{163}$$

Thus leapfrog has order two. Similarly, Simpson quadrature gives

$$\mathcal{L}_\tau[x](t_n) = -\frac{\tau^5}{90}x^{(5)}(t_{n+1}) + O(\tau^7), \tag{164}$$

so Simpson's multistep formula has order four.

18.4. **Algebraic order conditions.** Expanding (162) about $t = t_n$ gives

$$\mathcal{L}_\tau[x](t) = \gamma_0 x(t) + \gamma_1 \tau x'(t) + \gamma_2 \tau^2 x''(t) + \cdots,$$

where the scalar coefficients in this lecture are

$$\gamma_0 = \sum_{j=0}^{k} \alpha_j,$$
$$\gamma_m = \frac{1}{m!}\sum_{j=0}^{k} \alpha_j j^m - \frac{1}{(m-1)!}\sum_{j=0}^{k} \beta_j j^{m-1}, \qquad m \geq 1.$$

The notation γ_m distinguishes these scalar residual coefficients from the global-error functions $C_m(t)$.

Lemma 3 (LMM order conditions)**.** *The method has order at least p if and only if*

$$\boxed{\sum_{j=0}^{k} \alpha_j = 0, \qquad \sum_{j=0}^{k} \alpha_j j^m = m \sum_{j=0}^{k} \beta_j j^{m-1}, \quad m = 1, \dots, p.} \tag{165}$$

For $m = 1$, the last sum is simply $\sum_j \beta_j$.

Proof. The conditions are exactly $\gamma_0 = \cdots = \gamma_p = 0$, which are sufficient by Taylor's formula. Necessity follows by testing successively on constant, linear, and higher-degree polynomials. □

The first two conditions have the particularly useful form

$$\boxed{\rho(1) = 0, \qquad \rho'(1) = \sigma(1).} \tag{166}$$

An LMM satisfying these conditions is called *consistent*; it has order at least one. Equivalently, the general order conditions can be encoded as

$$\rho(e^z) - z\sigma(e^z) = O(z^{p+1}) \qquad (z \to 0).$$

This follows by inserting the exponential test function into the linear residual operator.

Before normalization there are $2k+2$ coefficients; an overall nonzero scaling leaves the method unchanged, so there are $2k+1$ effective parameters. Comparing this number with the $p+1$ order conditions suggests the formal upper bound $p \le 2k$ discussed in class. To justify the bound, suppose a nontrivial formula were exact on all polynomials of degree at most $2k+1$. Hermite interpolation at the $k+1$ distinct nodes allows arbitrary values of both a polynomial and its derivative at those nodes. The residual could vanish for all those choices only if every α_j and β_j were zero, contradicting $\alpha_k \neq 0$. This formal bound imposes no stability condition. The next lecture shows explicitly why even a consistent method of high order may fail to converge.

19. October 7, 2026

Consistency measures how well exact samples satisfy a multistep formula. To obtain convergence, we must also control how the recurrence

propagates perturbations. This leads to zero-stability and the root condition. We continue to use a uniform grid, $t_n = t_0 + n\tau$, and the polynomials ρ and σ from (160).

19.1. **A consistent method that is unstable.** Consider the explicit two-step formula from class,

$$x_{n+2} + 4x_{n+1} - 5x_n = \tau(4f_{n+1} + 2f_n). \tag{167}$$

Its characteristic polynomials are

$$\rho(\zeta) = \zeta^2 + 4\zeta - 5 = (\zeta - 1)(\zeta + 5), \qquad \sigma(\zeta) = 4\zeta + 2.$$

Since $\rho(1) = 0$ and $\rho'(1) = 6 = \sigma(1)$, the method is consistent. In fact, the order conditions for $m = 2, 3$ hold:

$$4 + 2^2 = 2(4),$$
$$4 + 2^3 = 3(4).$$

At $m = 4$, however, $4 + 2^4 = 20 \neq 16 = 4(4)$. Thus its order is exactly three, with residual

$$\mathcal{L}_\tau[x](t_n) = \frac{1}{6}\tau^4 x^{(4)}(t_n) + O(\tau^5).$$

Now apply it to $x' = 0$. The exact solution is constant, but the numerical recurrence is

$$x_{n+2} + 4x_{n+1} - 5x_n = 0.$$

Trying $x_n = \zeta^n$ gives $\rho(\zeta) = 0$, so every solution is of the form

$$x_n = a + b(-5)^n. \tag{168}$$

The constant mode is correct; the second mode magnifies perturbations exponentially.

For a concrete failure of convergence, take the IVP $x' = 0$, $x(0) = 0$, and starting values $x_0 = 0$, $x_1 = \tau^3$. These starting errors tend to zero at third order, but

$$x_n = \frac{\tau^3}{6}\big(1 - (-5)^n\big).$$

At fixed $T > 0$, with $n = T/\tau$, its magnitude grows without bound along the grids $\tau = T/N$. Small consistency error alone does not ensure convergence.

19.2. **The root condition and zero-stability.** Setting $f = 0$ in an LMM leaves

$$\rho(E)x_n = 0. \tag{169}$$

A simple nonzero root ζ of ρ produces a mode ζ^n; a root of multiplicity m produces modes

$$\zeta^n, \quad n\zeta^n, \quad \ldots, \quad n^{m-1}\zeta^n.$$

Roots strictly inside the unit disk give bounded, decaying modes, even when repeated. A zero root corresponds to a finite transient in the companion-matrix description. On the unit circle, any polynomial factor in n destroys boundedness.

Definition 18 (Root condition and zero-stability)**.** *An LMM is* zero-stable *if every root of ρ satisfies*

$$|\zeta| \le 1, \qquad \textit{and every root with } |\zeta| = 1 \textit{ is simple.} \tag{170}$$

This is the root condition*.*

Equivalently, there is a bound independent of the number of steps,

$$\sup_{n\ge 0} \|x_n\| \le K \max_{0\le j<k} \|x_j\|,$$

for solutions of the homogeneous recurrence (169). This uniformity matters because the number of steps to a fixed time tends to infinity as $\tau \to 0$. The term "zero" refers to the test equation $x' = 0$; absolute stability at nonzero $z = \tau\lambda$ will require a different polynomial.

Example 23 (A repeated unit root)**.** *If $\rho(\zeta) = (\zeta - 1)^2$, then*

$$x_{n+2} - 2x_{n+1} + x_n = 0, \qquad x_n = a + bn.$$

Thus $|\zeta| \le 1$ by itself is not enough. Starting values $x_0 = 0$, $x_1 = \tau$ yield $x_N = T$ at $N\tau = T$, despite both starting values tending to zero.

Example 24 (The methods from October 5)**.** *Leapfrog and Simpson's multistep formula have* $\rho(\zeta) = \zeta^2 - 1$*. Both roots,* 1 *and* -1*, are simple, so both methods are zero-stable. The* -1 *mode alternates without decay; zero-stability permits such a bounded parasitic mode.*

19.3. **Convergence and the Dahlquist equivalence theorem.**

Definition 19 (Convergence of an LMM)**.** *An LMM is convergent if, for every sufficiently regular IVP and every fixed finite interval* $[t_0, T]$*, starting values with*

$$e_{\text{start}}(\tau) := \max_{0 \le j < k} \|x_j - x(t_j)\| \longrightarrow 0$$

give numerical solutions satisfying

$$\max_{0 \le n \le N} \|x_n - x(t_n)\| \longrightarrow 0 \quad \text{as } \tau \to 0, \qquad t_N \le T.$$

Because the finitely many starting times approach t_0, the starting condition is equivalently $x_j \to x(t_0)$ for $j = 0, \dots, k-1$. Keeping $x(t_j)$ in the definition makes the required accuracy easier to quantify.

Theorem 11 (Dahlquist)**.** *A linear multistep method is convergent if and only if it is consistent and zero-stable, under the usual smoothness and Lipschitz assumptions on the IVP.*

More quantitatively, for a zero-stable method of consistency order p, sufficiently small τ gives an estimate of the form

$$\boxed{\max_{0 \le n \le N} \|x_n - x(t_n)\| \le C e^{L_*(T - t_0)} \big(e_{\text{start}}(\tau) + \tau^p\big).} \tag{171}$$

The constants are independent of τ, but may depend on the method, the interval, and bounds on the solution and f. In particular, order-p convergence requires starting errors of size $O(\tau^p)$.

To see the mechanism, put $e_n = x_n - x(t_n)$. Subtracting the exact residual equation from the numerical formula gives

$$\sum_{j=0}^{k} \alpha_j e_{n+j} = \tau \sum_{j=0}^{k} \beta_j \big(f(t_{n+j}, x_{n+j}) - f(t_{n+j}, x(t_{n+j}))\big) - \mathcal{L}_\tau[x](t_n). \tag{172}$$

The root condition bounds propagation by the homogeneous recurrence. The Lipschitz condition controls the first term on the right, and a discrete Gronwall argument gives a bound in terms of

$$e_{\text{start}}(\tau) + \sum_{n=0}^{N-k} \|\mathcal{L}_\tau[x](t_n)\|.$$

Since each residual is $O(\tau^{p+1})$ and there are $O(\tau^{-1})$ steps, their sum is $O(\tau^p)$. This is the multistep counterpart of the local-to-global argument for one-step methods on August 31.

For an implicit formula, the new value solves

$$x_{n+k} = \text{known terms} + \tau \frac{\beta_k}{\alpha_k} f(t_{n+k}, x_{n+k}).$$

For example, a Lipschitz constant L gives a contraction when

$$\tau \left| \frac{\beta_k}{\alpha_k} \right| L < 1.$$

This supplies a sufficient small-step solvability condition in the convergence argument.

19.4. **Adams methods.** The Adams family uses only the difference between the two newest solution values on the left:

$$x_{n+k} - x_{n+k-1} = \tau \sum_{j=0}^{k} \beta_j f_{n+j}. \tag{173}$$

Thus

$$\rho(\zeta) = \zeta^k - \zeta^{k-1} = \zeta^{k-1}(\zeta - 1). \tag{174}$$

The root at one is simple; all remaining roots are zero. Every method with this left-hand side is therefore zero-stable, independent of the derivative weights. Consistency additionally requires

$$\sigma(1) = \rho'(1) = 1, \qquad \text{that is,} \qquad \sum_{j=0}^{k} \beta_j = 1.$$

The usual Adams weights are obtained by integrating an interpolant of the derivative values. Excluding the new endpoint derivative gives

Adams–Bashforth; including it gives Adams–Moulton. Their construction and their nonzero-z stability are the topics of the next lecture.

20. October 9, 2026

We construct Adams–Bashforth and Adams–Moulton methods by integrating polynomial interpolants. We then distinguish their automatic zero-stability from absolute stability on $x' = \lambda x$. As before, $t_n = t_0 + n\tau$, $f_n = f(t_n, x_n)$, and $\tau > 0$ is constant.

20.1. **Adams formulas as quadrature rules.** An Adams method has the form

$$x_{n+k} - x_{n+k-1} = \tau \sum_{j=0}^{k} \beta_j f_{n+j}. \tag{175}$$

Its first characteristic polynomial is

$$\rho(\zeta) = \zeta^{k-1}(\zeta - 1).$$

Hence it satisfies the root condition for every k: one is a simple root, and all other roots lie strictly inside the unit disk. This proves zero-stability, regardless of the quadrature weights.

The exact solution satisfies

$$x(t_{n+k}) - x(t_{n+k-1}) = \int_{t_{n+k-1}}^{t_{n+k}} x'(t)\, dt.$$

Thus its residual in (175) is the quadrature error

$$\mathcal{L}_\tau[x](t_n) = \int_{t_{n+k-1}}^{t_{n+k}} x'(t)\, dt - \tau \sum_{j=0}^{k} \beta_j x'(t_{n+j}). \tag{176}$$

This is the same interpolation-and-integration principle used for collocation, but most of the derivative samples now come from previous steps.

Normalize time relative to the new endpoint:

$$t = t_{n+k} + \tau\theta, \qquad -1 \le \theta \le 0, \qquad v(\theta) = x'(t_{n+k} + \tau\theta).$$

The sample at t_{n+j} has node $\theta_j = j - k$. Given a set J of sample indices, form the Lagrange basis

$$\ell_j(\theta) = \prod_{\substack{m \in J \\ m \neq j}} \frac{\theta - \theta_m}{\theta_j - \theta_m}, \qquad j \in J.$$

Integrating the interpolant gives the weights

$$\boxed{\beta_j = \int_{-1}^{0} \ell_j(\theta)\, d\theta \quad (j \in J),} \tag{177}$$

with zero weights for omitted samples. In particular, the weights sum to one because the basis functions sum to one. This verifies the consistency condition $\sigma(1) = \rho'(1) = 1$.

20.2. **Adams–Bashforth: extrapolating known derivatives.** For the k-step Adams–Bashforth method (ABk), use only $j = 0, \dots, k-1$, so the nodes are $-k, \dots, -1$. The interpolant has degree at most $k-1$, and $\beta_k = 0$. The formula is explicit because all derivative values are known. Its use over $[-1, 0]$ extrapolates the derivative beyond the last known sample.

The interpolation error contains k time differences and the integration contributes one more power of τ. Therefore

$$\mathcal{L}_\tau[x](t_n) = O(\tau^{k+1}).$$

The order is exactly k: for the first omitted monomial, the interpolation remainder is proportional to $\prod_{m=1}^{k}(\theta + m)$, whose integral on $[-1, 0]$ is strictly positive. Since Adams methods are zero-stable, ABk converges with order k when initialized accurately enough.

The first three formulas from class are

$$\text{AB1:} \qquad x_{n+1} - x_n = \tau f_n, \tag{178}$$

$$\text{AB2:} \qquad x_{n+2} - x_{n+1} = \tau \left(\frac{3}{2} f_{n+1} - \frac{1}{2} f_n \right), \tag{179}$$

$$\text{AB3:} \qquad x_{n+3} - x_{n+2} = \tau \left(\frac{23}{12} f_{n+2} - \frac{16}{12} f_{n+1} + \frac{5}{12} f_n \right). \tag{180}$$

AB1 is forward Euler. For AB2, the basis functions at the nodes $-2, -1$ are $\ell_0(\theta) = -\theta - 1$ and $\ell_1(\theta) = \theta + 2$; their integrals are $-1/2$ and $3/2$, respectively.

For comparison with the fourth-order stability curve sketched in class, the same construction gives

$$x_{n+4} - x_{n+3} = \frac{\tau}{24}\big(55f_{n+3} - 59f_{n+2} + 37f_{n+1} - 9f_n\big). \tag{181}$$

The leading residuals, expanded about t_n, are

method	p	$\mathcal{L}_\tau[x](t_n)$
AB1	1	$\frac{1}{2}\tau^2 x''(t_n) + O(\tau^3)$
AB2	2	$\frac{5}{12}\tau^3 x'''(t_n) + O(\tau^4)$
AB3	3	$\frac{3}{8}\tau^4 x^{(4)}(t_n) + O(\tau^5)$
AB4	4	$\frac{251}{720}\tau^5 x^{(5)}(t_n) + O(\tau^6)$

20.3. **Stored derivatives and cost per step.** After initialization, ABk retains the latest k derivative values. It forms the new solution by a weighted sum, evaluates f once at that new solution, and discards the oldest derivative. Thus it needs only one new function evaluation per step, independent of k. For example, AB3 uses one new evaluation per step, whereas a standard three-stage third-order explicit RK method uses three. The saving comes with storage, startup requirements, and different stability restrictions. If the time step changes, the interpolation nodes and hence the weights generally change too.

20.4. **Adams–Moulton: including the new endpoint.** For the implicit Adams–Moulton construction in the indexing (175), take $J = \{0, \dots, k\}$: the nodes are $-k, \dots, 0$, and the interpolant has degree at most k. Now $\beta_k \neq 0$, so an implicit equation must be solved for x_{n+k}. The residual is $O(\tau^{k+2})$, and the method has order $k + 1$. The first omitted interpolation remainder has one sign on $(-1, 0)$, showing that this order is exact.

For example, nodes $-1, 0$ give the trapezoidal method

$$x_{n+1} - x_n = \frac{\tau}{2}(f_n + f_{n+1}), \qquad p = 2,$$

and nodes $-2, -1, 0$ give

$$x_{n+2} - x_{n+1} = \tau\left(-\frac{1}{12}f_n + \frac{8}{12}f_{n+1} + \frac{5}{12}f_{n+2}\right), \qquad p = 3. \tag{182}$$

Backward Euler is the first-order member of the Adams–Moulton family, using only the new endpoint derivative. Names such as "AM3" often refer to order rather than number of steps; here we state both the formula and its order to avoid this ambiguity.

An Adams–Bashforth prediction can provide an initial guess for the implicit solve. Using a finite number of correction sweeps instead of solving the Adams–Moulton equation defines a predictor–corrector algorithm whose stability must be analyzed as that actual algorithm.

20.5. **Absolute stability of a general LMM.** On the scalar test equation $x' = \lambda x$, with $z = \tau\lambda$, a general LMM becomes

$$\rho(E)x_n = z\sigma(E)x_n.$$

Trying a mode $x_n = \zeta^n$ gives the stability polynomial

$$\boxed{\pi_z(\zeta) = \rho(\zeta) - z\sigma(\zeta) = 0.} \tag{183}$$

For a genuine k-step recurrence we require $\alpha_k - z\beta_k \neq 0$, so the new value is uniquely determined from the history. Unlike the RK test equation, there are generally k amplification modes, rather than a single stability factor $R(z)$.

Following the convention used for RK methods, define

$$\mathcal{S} = \{z : \alpha_k - z\beta_k \neq 0, \text{ all roots of } \pi_z \text{ have } |\zeta| < 1\},$$
$$\mathcal{D} = \{z : \alpha_k - z\beta_k \neq 0, \pi_z \text{ satisfies the root condition}\}.$$

The first set gives decay of every numerical solution; the second gives boundedness for every initial history. At $z = 0$, this is exactly zero-stability. A consistent zero-stable method has a simple root at $\zeta = 1$, so $0 \in \mathcal{D}$ but $0 \notin \mathcal{S}$.

A candidate boundary is obtained by putting a root on the unit circle:

$$z(\theta) = \frac{\rho(e^{i\theta})}{\sigma(e^{i\theta})}, \qquad 0 \le \theta \le 2\pi, \tag{184}$$

where the denominator is nonzero. This is the boundary-locus construction. All the other roots must still be checked: a traced curve need not consist entirely of the boundary of $\mathcal{D}$.

20.6. **The stability restriction for AB2.** Substituting $f_n = \lambda x_n$ into AB2 gives

$$x_{n+2} - \left(1 + \frac{3}{2}z\right) x_{n+1} + \frac{1}{2}z\, x_n = 0.$$

Thus

$$\pi_z(\zeta) = \zeta^2 - \left(1 + \frac{3}{2}z\right)\zeta + \frac{1}{2}z, \qquad \rho(\zeta) = \zeta^2 - \zeta, \quad \sigma(\zeta) = \frac{3}{2}\zeta - \frac{1}{2}. \tag{185}$$

Its roots are

$$\zeta_\pm(z) = \frac{1 + 3z/2 \ \pm \sqrt{(1 + 3z/2)^2 - 2z}}{2}.$$

Near zero, one root approximates the physical mode $\zeta_+(z) = 1 + z + O(z^2)$, and the other satisfies $\zeta_-(z) = z/2 + O(z^2)$.

For real $z < 0$, the discriminant $1 + z + 9z^2/4$ is positive and the root product $z/2$ is negative. Write the roots as $\zeta_- < 0 < \zeta_+$. Since

$$\pi_z(1) = -z > 0, \qquad \pi_z(-1) = 2 + 2z,$$

the positive root lies below one, and the negative root lies above minus one exactly when $z > -1$. Hence

$$\boxed{\mathcal{S} \cap \mathbb{R} = (-1, 0), \qquad \mathcal{D} \cap \mathbb{R} = [-1, 0].} \tag{186}$$

For $z > 0$, $\pi_z(1) < 0$ forces a root larger than one. At $z = -1$, the roots are -1 and $1/2$: the iterates are bounded, but an alternating component does not decay. For a negative real λ, strict decay therefore requires $0 < \tau < 1/|\lambda|$.

The same polynomial test yields the negative real intervals

method	$\mathcal{S} \cap \mathbb{R}$	$\mathcal{D} \cap \mathbb{R}$
AB1	$(-2, 0)$	$[-2, 0]$
AB2	$(-1, 0)$	$[-1, 0]$
AB3	$(-6/11, 0)$	$[-6/11, 0]$
AB4	$(-3/10, 0)$	$[-3/10, 0]$

.

In each case the left endpoint is reached when a root passes through $\zeta = -1$, so it equals $\rho(-1)/\sigma(-1)$. The other roots satisfy the root condition there. These intervals illustrate the shrinking tolerance for negative real modes as the AB order increases. Zero-stability of every Adams method guarantees control in the small-step convergence limit; it does not remove the absolute-stability restriction at a chosen finite step.

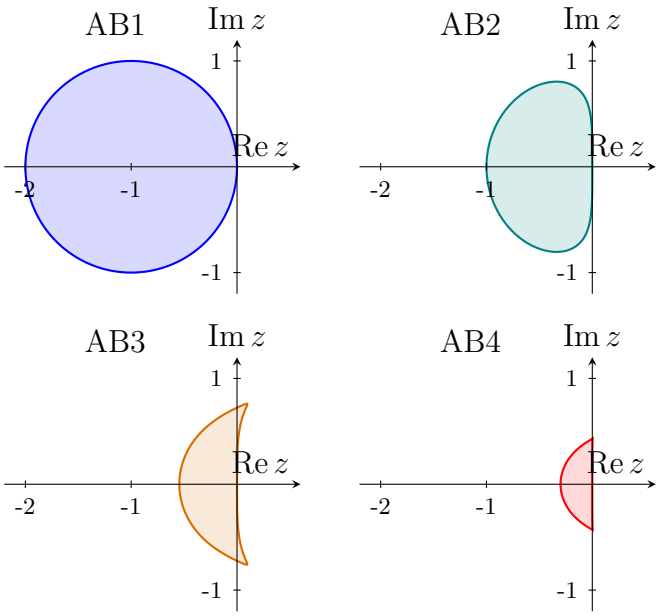


FIGURE 13. Computed stability domains for the first four Adams–Bashforth methods on the same scale. Shading indicates the root condition. For AB4, only the stable parts of the boundary locus are retained.